Ingredient mixing

A robot environment for mixing food by hand tool, hand mixer and on the hob

Gym-Anything for Robotics, environment `ingredient-mixing`.



The station as built and rendered: the kitchen, the two arms on their cart at the counter, the task's ingredients in the bowl and what they were measured from beside it.

Environment `ingredient-mixing` of Gym-Anything for Robotics. Action group "mix ingredients" (46 source rows). This report states what was built, on what basis, what was measured, and what is not right yet. Every number in sections 11 and 12 is read from a recorded run or bench. Nothing here was calibrated against a physical trial.

The mixture's counter is the rigid world's counter-top box, and its floor is the rigid world's ground plane. The records quoted here were made with an unbounded plane at the counter top's height in place of the box. Over the counter, where every task's grid region lies, the box and the plane give the same signed distance, normal and solid share to rounding, and the sauce task's rest and tilt runs read the same with either. A batch thrown into the air is drawn with merged droplets when one droplet per point would exceed the renderer's 60 MiB message.

01 / Summary

What the environment is, what was seen done in it, and what is not right.

The scene. A teaching kitchen 5.4 m by 4.2 m with daylight from a window: a quartz counter 0.914 m high along a tiled wall, a cart with two Franka arms and Franka Hands docked at it (FR3 by default; Pandas, and the right arm alone, can be chosen), a prep table, a storage rack, a hand-wash basin, a lesson board that carries the task. On the counter: a 5-quart steel mixing bowl (or a 1.75 L glass bowl, or a 1.5-quart saucepan on a two-zone induction hob), a ceramic crock with a balloon whisk, a silicone spatula and a wooden spoon (on the hob: a flat stirrer and a probe thermometer), a hand mixer standing on its heel, and the packs the recipe was measured from, each a free body with its own mass (section 3).

The food is an explicit material-point continuum on the GPU, 105,000 to 617,000 points, each carrying the mass of every ingredient it holds (section 4, 5). Its laws come from published measurements where one was found and are labelled where none was (section 8). A whisk's wires and a beater's strips are thinner than the solver's 2.5 mm cell; what such a wire does to powder beside it, and the liquid it lets into the powder it cuts, are read off a wire the same solver resolves on cells of 0.2 to 0.4 mm (sections 4 and 11.3). The hob heats the pan's base and the food through a film on the bottom, boils at the wall, evaporates at the surface and burns on when the film dries (section 7).

The tasks (section 10): dry ingredients whisked; a muffin batter folded; a pancake batter whisked until just combined; an

instant pudding whisked; a cake batter beaten with the hand mixer; a sauce of three pastes blended; a starch sauce cooked on the hob without catching; cold porridge reheated through. Each is judged on the food itself.

What was shown (sections 11.5 and 11.6):

Task	Assisted, inside all its limits (work until inside)	By the robot with the food (scripted operator)
`muffin-fold`	yes, 22 s, spatula to the batter's edge	yes: FR3 pair, right FR3 alone, right Panda alone
`dry-whisk`	yes, 98 s; in two of three processes	not run
`pancake-whisk`	yes, 38 s, with the whisk	no: FR3 pair, 0.142 against 0.10 after 44 strokes
`pudding-whisk`	yes, 44 s	not run
`cake-mixer`	yes, 30 s, beaters 5 mm over the floor	not run
`sauce-blend`	yes, 118 s, on the least even paste	yes: right Panda alone, Panda pair; no: right FR3 alone (0.107)
`stir-sauce`	yes, 298 s of stirring	no: Panda pair, two-handed, 110 cm2 burnt on
`reheat-porridge`	yes, 270 s	not run

What is not right (section 12 has every limit with its numbers): the robot, as scripted, has finished two of the eight tasks with the food (`muffin-fold`, `sauce-blend`), fails two (`pancake-whisk`, `stir-sauce`) and has not been run on the other four; one arm alone lets the bowl tip; a whisk takes a hundred seconds on dry powder where a cook takes ten, and where it meets the limit varies by run; the wire's rules stand on one resolved wire and the cell size still shows where they were not fitted; thin liquids are too thick and creep at rest (a trial at the cause failed); the burnt film, the film's thickness and much of the material's timing are estimates or transfers; no air; nothing compared with a physical trial.

02 / The activity and the scenario

The group's 46 rows describe one physical job done in many places: combine ingredients into a mixture with the properties the next step needs. They fall into mixing in a bowl with a utensil, whisking, doughs and batters, electric mixers, stirring cooking food, sauces, drinks, animal feed and a few outside a kitchen. The research found three jobs a cook names "mixing": *distributing* (dry into dry: a random mixture is the best possible end), *wetting and dispersing* (powder into liquid: lumps, dry pockets, and for wheat flour a limit on how much work the batter may take), and *homogenising while something changes* (heating, setting: the mixture's own properties move under the tool). The eight tasks cover each at least twice.

A teaching kitchen was chosen because one station sees all of it with the same tools, and because the arms' reach was measured for it before anything was built (bases 0.95 m high, 0.20 m in front of the counter edge: an FR3 holding a 0.29 m whisk reaches 38-43 of 45 tool-tip poses in the bowl).

03 / Scene

A teaching kitchen with one robot station. Whatever is drawn where an arm, a held tool or a thrown one can meet it has a collision proxy.

World frame: origin on the floor under the counter's front edge at the bowl station; x along the counter to the robot's right, y from the front edge to the wall, z up.

The room is one bay of a teaching kitchen, 5.4 x 4.2 x 2.7 m: fridge, sink unit, the counter run with an open shelf over it, a hand-wash basin at its right end, a window over a radiator with daylight from a pinned sky (Poly Haven "Kiara 5 Noon") and ceiling panels giving 500 lx at the vessel, the lesson board carrying the task's instruction, a prep table, a dry-store rack. The cart with the two arms is docked at the counter. **On the counter** beside the work stands what the task's recipe was measured from (flour, sugar, milk, eggs, the cake mix's box), each a free body with the mass of its pack (`params.PROP`: a 1 kg bag of flour 1.02 kg, a litre of milk 1.06 kg, six eggs in their tray 0.42 kg); an arm that sweeps one pushes it or knocks it over and the evaluator does not care.



Figure 2. The kitchen, from the door's corner and from the corner by the basin (the `cake-mixer` scene without the mixture). The lesson board carries the task's instruction.



Figure 3. The station. Left: the bake preset with `muffin-fold` in the bowl. Right: the hob preset with `stir-sauce` in the pan and the crock beside the hob.

Element	Geometry	Basis
Room	5.4 x 4.2 x 2.7 m; window 1.6 x 1.25 m in the right wall; door 0.92 x 2.04 m in the front wall	authored
Counter	top at 0.914 m, 0.635 m deep, 2.4 m long, 30 mm quartz	NKBA figures (search)
Cart	deck at 0.95 m; arm bases 0.56 m apart, 0.20 m in front of the counter edge, facing the wall	measured reach (M4)
Mixing bowl	45 mm flat then an arc to a 149 mm rim, 114 mm deep (5.3 L brimful); 0.55 kg	Vollrath 69050 (read)
Glass bowl	rim radius 100 mm, 81 mm deep, 0.60 kg	search; mass open
Saucepan	152 mm across, 89 mm deep, tri-ply, 190 mm handle, 0.91 kg	All-Clad D3 1.5 qt (search)
Crock	ceramic, 180 mm across inside, 165 mm deep, three-way divider, 1.6 kg	estimate
Whisk	292 mm; cage 175 x 64 mm of ten 1.4 mm wires; 0.10 kg	OXO 11.5 in (read); wire search
Spatula	279 mm; blade 60 x 100 mm silicone, sprung at mid-length	GIR 11 in (search); stiffness estimate

Element	Geometry	Basis
Spoon, stirrer	305 mm beech, 9.5 mm shank, bowl 51 x 79 mm, hollow 7 mm deep; 300 mm flat stirrer, 40 mm edge	outline/shank read; hollow is an estimate
Thermometer (hob tasks)	pen type, 235 mm, 0.03 kg; tip follows the food within 6 mm, 1 s time constant	estimate; time constant from search only
Hand mixer	1.0 kg, 188 x 88 x 110 mm housing, round bar handle, two beaters; stands on its heel	mass: KitchenAid KHM5AP specification sheet (2.3 lb, read); see section 6 for the beaters
Hob	zone at (0.75, 0.32); a knob on the counter in front with off and twenty detents 15 degrees apart	levels and wattages: the Duxtop manual (read); see section 6 for the knob's detent torque
Fittings, furniture	sink, towel rail, shelf canisters, prep table, stools, rack, basin, pedal bin, fridge, plant	authored; rack/stools/crates/plant are pinned Poly Haven models; static proxies, each struck (11.7)
On the counter	per task, three to seven packs, jars and tools of the recipe (`params.PROPS`)	authored outlines; free bodies with their packs' masses (11.7)

The crock stands 0.37 m to the right of the vessel, in front of the right arm, which takes the tool; the left hand is for the pan's handle, the rim, the knob or the mixer (on its heel 0.34 m to the left). Everything an arm, a held tool or a thrown one can meet collides; walls, ceiling and wall fittings are static proxies. Section 11.7 strikes each of the 39 free things on the counter and records how far it was pushed, and checks that all 39 stand where placed once a scene is built; the fridge and the plant's pot are solid too, the plant's leaves not.

Three choices were made because a simpler one failed a measured test: the crock's position relative to the pan, the open shelf instead of wall cabinets, and the main camera's wall bracket instead of a cart-mounted mast. The tools' handles are round, as sold (a squared-sleeve variant, `sleeves`, remains); the mixer stands on its heel, as sold, and reaching its handle is the arm's affair (section 11.5). Every vessel is described once, by its inner profile and wall: from that come its rigid pieces, the signed-distance table the mixture collides with, the drawn wall, and its volume.

04 / The mixture: method

Explicit MLS-MPM with APIC transfers on a uniform grid (Hu et al. 2018; Jiang et al. 2015), written for this package in Warp (`continuum.py`). Cell Δx = 2.5 mm; eight points per cell at the start; numerical sound speed c = 6 m/s; substep $\Delta t = 0.25\,\Delta x/\sqrt{c^2 + 4G/3\rho}$, 11 to 13 per 1 ms tick.

The grid's region is rebuilt every substep in a world-aligned box round the vessel as it stands at that tick, turning with it (`world._region_box`, `continuum.set_box`): a box that does not turn with the vessel loses mixture past 15-25 degrees of tilt. A point outside the region flies ballistically; inside it, a flying point becomes mixture again at a solid, and sticks where it lands outside ("spilt"). Every point also carries the place and time at which it first left the vessel's own envelope, a cylinder from base to rim (`continuum.k_exit`, `measures.left_vessel`: above the rim, below it, or outside with no record); the envelope over-counts on a vessel narrower below its rim (51 g of 346 g on the 5 qt bowl's tilt recorded as if through the base, section 11.1, uncorrected in the evaluator).

Pressure. Each point carries a volume ratio J, drawn toward the ratio the gathered volume fraction implies:

$$J \leftarrow J\,(1 + \Delta t\,\mathrm{tr}\,L) + \big(1/\varphi_p - J\big)\,\Delta t/\tau_J, \qquad \varphi_p = \sum_i w_{ip}\Big(\sum_q w_{iq} V_q^0/\Delta x^3 + s_i\Big), \qquad p = K_b\,(1/J - 1)_+. \tag{1}$$

w are the quadratic B-spline weights, V^0 the points' rest volumes (section 5), s_i the solid's share of node i's weight, τ_J = 0.1 s, $K_b = \rho c^2$ for wet states. Next to a solid, where the solid's share exceeds 2% and shear is below 1 s^{-1}, the second term is switched off (otherwise a resting liquid circulates along the wall indefinitely; section 11.1, Figure 10). At a free surface φ_p is low because part of the kernel is in the air; the gradient $g_p = \Delta x\,|\nabla\varphi|_p$ tells how much, and for material at rest the fraction is corrected against what a resting lattice shows at that gradient:

$$\hat\varphi_p = \frac{\varphi_p}{H(g_p)}, \qquad H:\ (g, H) = (\le 0.07,\ 1),\ (0.367,\ 0.864),\ (\ge 0.596,\ 0.620), \tag{2}$$

linear in between, read from the solver itself; J is drawn toward $1/\hat\varphi_p$. Off while the material flows (shear above 1 s^{-1}) and next to a solid.

Deviatoric stress. With $D' = \mathrm{dev}\,\frac{1}{2}(L + L^{\mathsf T})$, $\dot\gamma = \sqrt{2D' : D'}$ and $\eta_a = K\dot\gamma^{\,n-1}$, a share $r = \min(1, 0.05\,\rho\,\Delta x^2/(\Delta t\,\eta_a))$ of the viscous stress is applied explicitly; the rest is carried by an elastic trial stress relaxed onto the flow surface:

$$s^{\mathrm{tr}} = s + \Delta t\,(2GD' + Ws - sW), \quad \tau^{\mathrm{tr}} = \sqrt{\tfrac{1}{2}s^{\mathrm{tr}} : s^{\mathrm{tr}}}, \quad \tau^{\mathrm{tr}} - G\,\Delta t\,\dot\gamma_p = \tau_y + \mu\,p + (1-r)\,K\,\dot\gamma_p^{\,n}, \quad s = s^{\mathrm{tr}}\,\frac{\tau^{\mathrm{tr}} - G\,\Delta t\,\dot\gamma_p}{\tau^{\mathrm{tr}}}. \tag{3}$$

$\mu = 0$ for a paste (Herschel-Bulkley); $K \approx 0$, $\mu = \sin 37°$ for a powder (Drucker-Prager with cohesion). G = 2 kPa (pastes) and 4 kPa (powders) are numerical, set so the elastic strain at yield is 1-5%.

Solids on the grid (`k_grid`, `k_wall`): a signed-distance solid per tool and vessel, moving with its rigid body. Three rules: (1) at every node within one cell of a solid, the approaching normal velocity is removed, and both ways once the material is pressed against it; (2) a viscous wall law holds a thick paste as before and lets a thin liquid slide, slowed only as fast as its own viscosity allows (constant c_w = 8, calibrated on the cup's laminar Couette torque), with two further rules of no fitted constant (`k_keep_out`, a point inside a solid set back on its surface each tick; a corner rule stopping a slide into a face turned more than 45 degrees from the current one) that keep a liquid from walking into a square inside corner, without curing a thin liquid's slow creep at rest (section 11.1); (3) dry powder gets Coulomb friction on the normal impulse (10.5 degrees on steel). A trial rule to cure that creep (`wall_thrust`, section 8) cured it at rest but broke the stirred cup's torque balance, so it is not used.

Thin wires (whisk loops, beater strips), thinner than a cell: a chain of 5 mm segments exchanging momentum with the grid by the drag of a cylinder at the local relative velocity U_n, plus the plastic resistance of a material with strength ($N\,s\,d$ per length, N = 10.5, Randolph and Houlsby 1984):

$$\frac{f}{\ell} = \frac{4\pi\,\eta_a U_n}{\frac{1}{2} - \gamma_E - \ln(\mathrm{Re}/8)}\ \ (\mathrm{Re} < 1), \qquad \frac{f}{\ell} = \tfrac{1}{2}\rho U_n^2 d\,\big(1 + 10\,\mathrm{Re}^{-2/3}\big)\ \ (\mathrm{Re} \ge 1), \tag{4}$$

half the coefficient along the wire, applied as an exponential relaxation toward the wire's velocity.

What a wire does to dry powder and stiff paste beside it, on the powder's own points, is three parts (`continuum.k_wire_sweep`). The first part alone left the whisk three times slower than a blade on the same strokes. *Out of the wire's way*: a point inside the wire's cylinder is set on its surface and takes the wire's speed there if slower. *Carried along* (`WIRE_CARRY`, share 0.5, decay length 0.7 wire diameters): a point outside the wire still goes its way at no less than that share of its speed, decaying with distance, read off a wire RESOLVED by the same solver on 0.2-0.4 mm cells through the same flour: the resolved wire carries 10.4 to 10.5 mm^2 of powder per length, the rule 8.4 and 8.1 on the tasks' cells and on cells half that size, where the first part alone carries 0.27 and 3.5; it acts on points, so its result does not hang on the cell in that bench (section 11.3 has where it still does). *The cut* (`WIRE_CUT`, 0.5 wire diameters of open volume per swept area): the resolved wire's own track holds only 0.43-0.50 of the powder it held. The drag law's own consistency, flow index and strength round a wire are means of the points' values, gathered once a tick and divided by the mass gathered with them (`grm`); divided by a later substep's mass they give a drag of infinity over infinity next to a drop in flight.

A cut the wire leaves is refilled by liquid that reaches it (`Continuum._slit`: `k_slit_scan/hop/sum/move`). A cut bordering free thin liquid fills from it and runs on along the cut: `CUT_REACH_D` = 6 wire diameters through dry powder, `CUT_PASTE_COST` = 1.7 cells of reach per cell through a paste holding at least `CUT_HOLD_PA` = 50 Pa (estimate), from free liquid no thicker than `CUT_THIN_PA_S` = 1 Pa s^n. The reach is measured on the same resolved wire drawn from a pool of thin liquid into a bed of flour: 6.0 mm2 of liquid per length of wire went into the bed with it (3.05 wire diameters squared; the rule gives 3.0), the same on two cell sizes and for a cut twice as long, and 3.5 mm2 into a stiff paste (1.80 diameters squared; the rule 1.76) (`CUT_*` in `ledger_mixture.py`; section 11.3). A heap free to give way takes more than a bed: the rule takes the bed's figure, and understates what a whisk does to a small or loose heap. The cut also counts how much liquid a stroke has taken into the flour, reported by the end-state runs as `liquid_into_cuts_ml`. The liquid is conserved to rounding, drawn pro rata from every point holding free thin liquid.

The film a wire carries (`continuum.k_film`, once per tick): a wisp (gathered fraction under 0.35) within 0.8 of a cell of a wire out of the mixture rides it; the film's thickness follows Derjaguin's thick-film limit $h = \sqrt{\eta U/(\rho g)}$, at most three quarters of the wire's width, drains under gravity less the wire's own acceleration, and sheds drops with chance Δt over the run-off time; a drop flies free for 0.3 s before a wire may take it again.

Coupling to the rigid world (`world.py`): each tick the mixture receives every body's pose and twist, advances its substeps, and returns the mean wrench. A light vessel holding heavier contents is unstable under naive staggered coupling (Causin, Gerbeau and Nobile 2005), so the contents' mass, centre of mass and transverse inertia are lumped into the vessel's rigid body every tick and the rigid solver gets the residual, low-pass filtered at 8 ms; the robot's fingers and hands are solids for the mixture too.

05 / The mixture: what a point is made of

Species (kg per point): water phase, fat, sugar, a minor powder, the hydrating solid of the recipe in four states (dry, wetted and lumped, suspended, cooked), and up to three finished pastes.

Volume. With particle densities ρ_k and, for each solid, the liquid it holds as a void ratio e_k:

$$V_s = \sum_{\text{solids}} \frac{m_k}{\rho_k}, \quad V_{\text{pore}} = \sum_{\text{solids}} e_k \frac{m_k}{\rho_k}, \quad V_w = \sum_{\text{liquids, pastes}} \frac{m_k}{\rho_k}, \qquad V^0 = V_s + \max(V_{\text{pore}}, V_w), \quad S = \min(1, V_w/V_{\text{pore}}). \tag{5}$$

Flour: 1470 kg/m^3 true (a search summary, not confirmed on an opened page), e = 1.88 dry (510 kg/m^3 poured) and 0.9 once wetted (0.88 ml of water per ml of solid, quoted from memory of farinograph figures). A rule that kept a wetted solid's poured pore volume left a muffin batter's last sixth of flour dry after any amount of folding.

States. Below S = 0.9 a granular material: friction 37 degrees, cohesion 80 Pa plus a capillary term peaking at 500 Pa near S = 0.6, bulk stiffness 6.2 kPa. At S = 1 a paste, blended over 0.9 < S < 1 so no property jumps. Drawing only (points, volumes and what the evaluator reads are untouched): a stirred powder drawn with a liquid's own sphere rule reads as curd, not flour, so a powder point is drawn with a larger sphere and extra smoothing (`surface.reconstruct(powder=...)`, `POWDER_RADIUS`, `POWDER_FAIRING`), any point whose neighbours stand apart is drawn larger in proportion so a clump the whisk pulled loose reads as a clump (`k_apart`, `APART_MOST`), and pits up to two cells across are closed (`CLOSE_PITS`). Spilt material on the counter is drawn with a round Gaussian kernel (`render._spill_grid`), since a binomial blur's square outline draws it as pale square flecks.

Batter law. With x the wetted-solid mass fraction,

$$\log_{10} K = a\,x + b, \qquad n = 1 - c\,x \ (\geq 0.3), \qquad \tau_y = 300\,\text{Pa}\left(\frac{x - 0.30}{0.31}\right)^2 \ (0.30 < x < 0.61), \tag{6}$$

a = 12.2, b = -3.4, c = 1.21 for wheat flour: an authored interpolation through the measured wheat pancake batter, the muffin batter and bread dough at farinograph absorption (Lorente-Bailo et al.; Ozuna et al.). Other solids replace the constants (whole spelt, cake mix, instant pudding, corn starch).

Finished pastes carry their own (τ_y, K, n); where they meet, Nagasawa et al.'s blending map sets the mixed point's parameters (mayonnaise and honey each resist more than their 1:1 mixture, which is kept), read off that paper's published flow curves by eye (mayonnaise 60 Pa, 40 Pa s$^{0.40}$; ketchup 25 Pa, 5 Pa s$^{0.44}$; honey 20 Pa s$^{0.95}$); relish, porridge and canned food are estimates.

Change with time, temperature and work. Hydration: $K \times (1 + 0.5\,(1 - e^{-t_w/300\,\text{s}}))$. Cold-set starch: a set fraction with a 100 s time constant taken from the packet's "soft-set in 5 minutes". Cooked starch: $k(T) = A\,e^{-E_a/RT}$, E_a = 150 kJ/mol, set so a 5 K/min ramp is 18% cooked at 60 C and 95% at 77 C; consistency gains $2.56 \times 10^4\,x_c^{2.73}$ Pa s^n. Temperature: $K \times \exp[(E_v/R)(1/T - 1/298)]$, E_v = 18.4 kJ/mol. Work: flour batter stiffens by $1 + (1 - e^{-w/w_d})$, w_d set so 25 reference strokes add a tenth (0.33 J/kg per stroke, measured on the method's own strokes, section 11.2).

Wetting, lumps, mixing below the points' spacing (`k_pool_*`, `k_refresh`): a conservative pool exchange on the grid, each point trading a fraction $r_p\,\delta t$ of every species with its neighbourhood:

$$r_p = \frac{D}{\ell^2} + \min\Big(\frac{D}{\ell^2}\big(e^{2\Lambda} - 1\big),\ c_m\,\dot\gamma\Big) + k_w\,\chi_w, \qquad \dot\Lambda = -\,n \cdot D' n, \quad \dot n = -L^{\mathsf T} n + (n \cdot D' n)\,n. \tag{7}$$

χ_w is 1 where free liquid and unfilled pores are both within reach and 0 elsewhere. Λ is the log-thinning of lamellae inside a point (folding thins them, a reversed shear un-thins them); D is molecular (dissolved species) or shear diffusivity (grains); wetting uses $k_w = D_w/\ell^2$ and stops once neither free liquid nor unfilled pores remain within reach, leaving a dry core under a wet skin. D_w = 2e-6 m2/s is calibrated on the muffin method's "15 to 20 light strokes" (Quaker Oats, Baking 101); no direct measurement of flour's own wetting diffusivity was found (section 11.2 has the bound from Washburn's law). The thinned share is bounded by $c_m\dot\gamma$ (a borrowed, uncalibrated sub-grid diffusivity) so a strained-then-stopped material does not go on exchanging at the thinned rate.

A point whose saturation passes 0.5 turns its dry solid into lumped solid: a modelling choice, stated as one (how the powder meets the liquid cannot matter here, only the working afterwards). Lumps erode into suspension at

$$\dot\lambda = -k_e\,\dot\gamma\,\lambda\,\max(0,\ 1 - \tau_c/\tau), \qquad k_e = 0.05,\ \tau_c = 20\,\text{Pa}, \tag{8}$$

(Rwei, Manas-Zloczower and Feke 1991), and additionally where a wire sweeps, by the same law over a strain of 2; both constants are estimates chosen between what the tools measurably do (section 11.2).

06 / Tools, vessels and appliances

- **Handles** (`tools.handle`): every hand tool has the round handle it is sold with (whisk 28 mm, spatula 16 mm, spoon's shank 9.5 mm, stirrer 12 mm, thermometer 16 mm); the `sleeves` variant carries square-section sleeves instead, and says so in its instruction.
- **Whisk**: handle and wire cage as one body, sprung 440 N/m at the tip (ten 1.4 mm wires as 170 mm cantilevers, 44 N/m each); four spheres for rigid contact, ten wire chains for the mixture (section 4).
- **Spatula**: the blade's outer half is on a sprung hinge (250 N/m at the tip, estimate); for the mixture a blade thinner than two cells is drawn two cells thick.
- **Spoon** (`tools.Paddle`, `tools.dish_boxes`): rigid, its bowl a dish of 14 boxes holding 16 ml level; the hollow exists physically (shown with a pea and a cube, section 11.3), but no probe shows it lifting a spoonful.
- **Stirrer**: rigid, a flat blade with a 40 mm edge.
- **Probe thermometer** (`tools.Thermometer`): reads the mean temperature within 6 mm of its tip, 1 s time constant, drifting to the room's 20 C with 15 s in air (section 11.4); its 3.5 mm stem is not felt by the mixture.
- **Hand mixer** (`tools.Mixer`): stands on its heel, handle a vertical round bar, beaters pointing down; the slide tab releases at `MIXER_DETENT_FORCE` = 3 N, bounded by a slide-switch data sheet (1.5-3.4 N over a 3 mm stroke). The five speeds (`MIXER_SPEEDS`) are 0, 288, 493, 698, 902, 1107 rpm, the lowest and highest read from a hand-mixer review's own measurements (America's Test Kitchen, KHM512), the three between even steps (derived). Beater radius (`MIXER_BEATER_R`) is 22.2 mm, read from the same review's "Beater Head Diameter: 1.75 in". Stall torque (`MIXER_STALL`) is an estimate, 1.4 N·m, derived from a maker's data sheet (85 W at 580 rpm, KitchenAid 5KHM5110).
- **Hob** (`heat.py`, section 7): a rotary knob, off and twenty detents 15 degrees apart. Detent torque (`HOB_DETENT_NM`) is 0.035 N·m, the top of a detented-knob maker's range for more than 13 detents (Piher PTC-15 data sheet). The positions are the Duxtop manual's twenty power levels, 100-1800 W; `stir-sauce` starts at the twelfth (1000 W), `reheat-porridge` at the seventh (580 W). Pan detection and an overheat cut-out as in section 7; a hand turns the knob (section 11.3).



Figure 4. The tools and the appliance as the policy's cameras see them. Left, the station camera in the bake preset: the crock with the whisk, the spatula and the spoon on their round handles, the steel bowl, the hand mixer standing on its heel, the arms' bases beyond. Right, the right wrist camera at the ready pose, over the crock.

07 / Heat (preset hob)

Points carry temperature; heat moves between them by the pool exchange at the mixture's diffusivity. The pan's base is a plate of 96 patches (8 rings by 12 sectors) with the tri-ply's heat capacity and in-plane conduction; the wall is one more node losing heat to the room. The hob's power is deposited in a ring of mean radius 45 mm, only while the pan stands on the zone, and cuts out above 238 C at a sensor that follows the pan's base with an 8 s lag (the Duxtop manual's protections; section 11.4).

Between a patch at T_w and the food above it there is a film thinner than a cell:

$$q = h\,(T_w - T_b), \qquad h = \sqrt{\frac{k\rho c_p}{\pi\, t_r}}, \qquad \dot{t}_r = 1 - \dot{\gamma}\, t_r, \tag{9}$$

penetration theory, t_r the time since the patch was renewed (shear in the food on it renews it). A wet patch may run above 100 C and pass heat by nucleate boiling as well:

$$q_b = B\,(T_w - 100)^3, \qquad B = 137\ \mathrm{W\,m^{-2}K^{-3}}, \tag{9b}$$

Rohsenow's correlation for water on polished stainless steel; `HOB_BOIL_B` is **read** (the correlation and surface constant reproduce a worked example's own answer and put this hob's plate 7.0-7.1 C over boiling at its own flux). Both exchanges are stiff and stepped linearised about the plate's temperature; vapour is made only where the food itself has reached 100 C.

A burnt-on patch's dried film sits between the plate and the food as a layer in series, of the film's remaining thickness and a fouling deposit's conductivity (`params.HOB burnt_k` = 0.37 W/m/K, **read**: a 15% non-fat-dry-milk fouling layer's conductivity in a heated stainless tube, Cantarero Rivera, Mishra, Ozadali and Benyathiar 2021, via Europe PMC). Conduction and boiling both proceed from the layer's top, so the plate under a burnt patch runs hotter by the layer's own temperature drop. A rule that let a burnt patch pass only 0.3 of the conduction figure and no boiling ran it to 249 C within a minute with the sauce stalled under it. A scraped burnt patch gives up flecks of its film as dark specks at the work of scraping, `HOB SCRAPE` = 10 J/m2 (read, whey-protein deposits on steel, transferred to a starch film). The film thickness that sets how soon an unstirred sauce catches, 0.15 mm, is an estimate; no measurement of the film or of a starch-and-milk scorch time was found (section 11.4 has what the model gives against one cook's unverified "15 seconds" claim, not changed to meet it).

The food's surface loses heat by evaporation. From 50 C up the loss is a measured pot's own: 5 litres of water in an open stainless pot on an electric range, held at each temperature, no lid (A. Berick, *Heat Losses In A Cook Pot at Constant Temperature*, Aprovecho Research Center, figure 5, read off the printed chart): straight lines between 74 g/h at 50 C and 1128 g/h at 98 C. Carrier's law gives 1.05 of the measured loss at 50 C and 0.59 at 98 C, so it is used below 50 C only, scaled by 0.95 to meet the table there (`evaporation_kg_s_m2`, `params.HOB CARRIER`). The mean food temperature below which vapour made at the pan bottom collapses rather than leaving, `HOB_SIMMER_C`, is a **read** 96 C: the same source's figure 7 gives a straight heating-curve slope, from an oiled pot that cannot evaporate from its surface, holding to about 96 C before the plateau, and a second source (bubbles reaching the surface at 5 K subcooling and not at 9.3 K or more, Isikan 1986, a thesis read only second-hand) points the same way.

Steam and simmer are drawn from the heat model's own rates (one puff per 4 mg/s leaving, one bubble per 25 W), not simulated as air or foam, read by no section 10 criterion; the vapour's mass is not taken from the sauce (it does not thicken here by reduction). The blended pastes' colour is drawn by a two-constant Kubelka-Munk rule, each non-mayonnaise paste scattering at 0.1 of mayonnaise's own (estimate; no direct measurement found). A liquid thinner than the solver's numerical floor (0.04-0.07 Pa s) cannot convect as it would in a pan; for it the film coefficient and inter-point conduction are raised by a natural-convection closure (McAdams), fading out as consistency rises past the floor — a labelled closure for a known solver defect, not a property of the food (section 11.4).

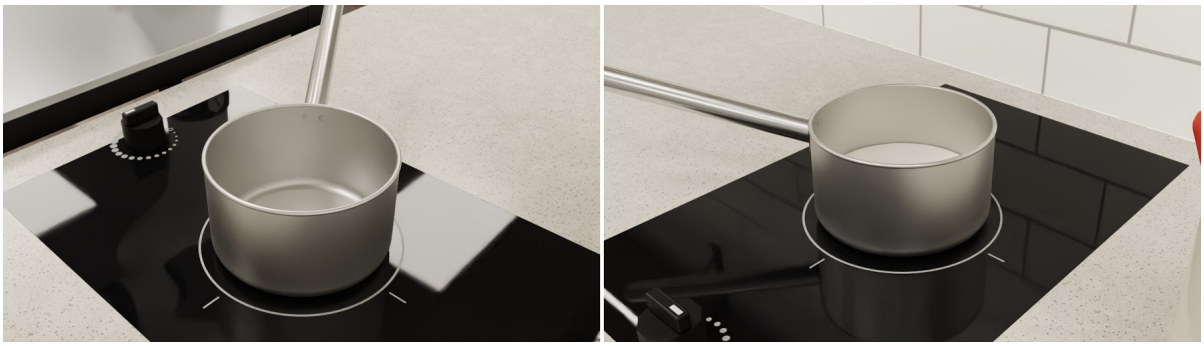


Figure 5. The hob. Left: the saucepan on its zone and the knob with its ring of marks. Right: the pan with the cold sauce at the start of `stir-sauce`.

08 / Parameters

`python -m ingredient_mixing.params` prints the ledger (value, unit, label, basis): **221 entries** (128 for geometry, robots and appliances in `params.py`, 93 for food in `ledger_mixture.py`). By label: 63 estimate, 31 repo, 25 read, 24 search, 22 authored, 18 literature, 12 engine, 11 measured, 8 derived, 4 open, 3 calibrated. The three calibrated entries (wetting diffusivity, wall law constant, wall rule distance) are each set on one observable and tested on others (section 11).

What	Value	Label
Pancake batter, wheat	K 0.754 Pa s^n, n 0.676, 1135 kg/m3, 3358 J/kg/K	read (Lorente-Bailo et al.)
Mayonnaise	78.8 Pa, 71.3 Pa s$^{0.3}$ measured; used: 60 Pa, 40 Pa s$^{0.40}$ inside the blend rule's range	read (Flamminii et al. 2020); graph
Flour	90 um; 510 kg/m3 poured, 880 at 4.5 kPa, 1470 true; friction 37 deg, wall 10.5 deg	estimate; search, not confirmed on an opened page; estimate
Wetting diffusivity	2e-6 m2/s	calibrated on the muffin method's 15-20 strokes (11.2); no flour measurement found
Wire carry / cut / slit reach	share 0.5, decay 0.7 d; 0.5 d open volume per swept area; cut reach 6 d, paste cost 1.7, hold 50 Pa, thin limit 1 Pa s^n	measured on a resolved wire; hold is an estimate
Film that dries before burning on	0.15 mm	estimate; no measurement of the film itself was found
Burnt-film conductivity	0.37 W/m/K	read (a dried dairy fouling layer; transferred to a starch film)
Evaporation from the food's surface	measured pot, 74-1128 g/h from 50 to 98 C; Carrier's law below 50 C only, x0.95	read (Berick, Aprovecho)
Simmer threshold	96 C	read (Berick's oiled-pot heating curve; a second bubble-subcooling source agrees)
Nucleate boiling constant B	137 W/m2/K3	read (Rohsenow's correlation)
Over-mixing reference stroke	0.33 J/kg	measured here on the method's own strokes (11.2)
Mixer speeds / beater radius / stall / detent force	0,288,493,698,902,1107 rpm; 22.2 mm; 1.4 N m; 3 N	read (two endpoints, a review); read; estimate (derived from a maker's wattage); estimate (bounded by a slide-switch data sheet)
Hob detent torque	0.035 N m	read (a detented-knob maker's range for >13 detents)
FR3	2.62-5.26 rad/s, 10 rad/s2, 5000 rad/s3, 87/12 N m, 70 N grasp, 3 kg; configured load at most 2.27 kg	repo (drilling ledger)
Evaluator limits	unevenness 0.10; dry flour 2%; lumps 2%; 25 strokes' work; 95% cooked and the sauce having reached 88 C; 2 cm2 burnt; every portion at 74 C or more within 15 K; 1% spilt; 50 mm	section 10 gives the basis of each
Numerics	cell 2.5 mm, c 6 m/s, G 2 and 4 kPa, τ_J 0.1 s, wall law constant 8, approach band 1 cell, rest stress 1 Pa, the free-surface table of section 4	engine; calibrated or measured choices (11.1)
Solver rules with no constant of their own	a point inside a solid is moved back onto its surface each tick (`k_keep_out`); a point sliding along one wall face is stopped at the next face past 45 degrees; dry powder inside a wire's cylinder is moved onto its surface and carried at the wire's speed (`k_wire_sweep`, part 1)	engine
Wall-thrust trial	an added node-force term that removed resting creep along a wall but broke the stirred cup's torque balance by 57%	engine, not used

09 / Robots and interface

Configurations (`robots/catalog.py`, the `ingredient-mixing` entry; `robot_setup.py`). The station takes fixed-base Franka arms with a Franka Hand each from the shared catalogue, on the cart's two mounts; nothing of an arm is written into the scene.

Name	Rig	What it is	Offered for
`dual-fr3` (default)	`mix-fr3`	two `fr3@1` arms, controller profile `barcode_scanning`	every task
`dual-panda`	`mix-panda`	two `panda@1` arms, controller profile `drilling` (35 N a finger)	seven tasks: every task but `cake-mixer`
`single-fr3`	`mix-fr3-single`	the right FR3 alone	`dry-whisk`, `sauce-blend`, `muffin-fold`, `pancake-whisk`, `pudding-whisk`
`single-panda`	`mix-panda-single`	the right Panda alone	the same five

`robot_setup.check_task` refuses a task at launch and at reset, with the reason and which configurations ARE offered, whenever one is not: the Panda pair is withheld from `cake-mixer` (a mixture-free rehearsal ended with the mixer running, unreachable by the hand that had just switched it on); a single arm is withheld from both hob tasks and from `cake-mixer`; the left FR3 alone is refused for every task (it cannot reach the whisk, lift the spoon clear of the crock, or both hold and switch off the mixer). `g1`, the Unitree G1 humanoid, was built and measured at this counter without the mixture and **is not offered for any task**: a 43-joint interface, a hand grasp command and a head camera are not built, and the arm controller, reflexes and load model remain the Franka arms'. `mobile-fr3-duo/-single`, `panda`, `dual-panda-wsg50` and `yam` are refused too, each for its own reason (section 11.8).

The arm controller is drilling's: a time-synchronised jerk-limited trajectory (Ruckig) within 95% of the Panda's own velocity limits; desired torque capped at the joint ratings, weight compensated outside the cap; a collision reflex (external joint torque less the configured load, filtered at 20 ms) stops an arm for 1 s. The Franka Hand closes at 50 mm/s until blocked, then holds the commanded force (up to 35 N a finger). **The load is configured, not guessed** (`World.set_load`, libfranka's `setLoad`): a policy or the scripted operator tells the controller what mass and centre of mass the hand carries, at most 2.27 kg; a hand that only steadies something standing on its own sets no load, and the mixture's drag on a held tool is never part of it (it shows up in the measured joint torques instead).

The hand holds a place (`world._pad_stick`, `params.PAD_STICK`). Three rules hold it: the two fingers of a Franka Hand are tied stiffly as one mechanism; a pad's friction removes a slip speed in one tick instead of at a damping rate; each pad contact springs back to the place it gripped, up to an elastic reach of 0.2 mm, as rubber in shear does. One known error remains: two contacts between a pad and a crossing bar resist twisting as well as the torsional row does, holding up to three times a real pad's twist.

The scripted operator and the Panda's hand. An operator that streams a new joint target every 20 ms whatever the arm does leaves the arm a median 10 mm (at most 33 mm) behind the streamed pose, on the Panda and on the FR3 alike. Run that way, ten strokes of `dry-whisk` with the mixture left the spatula's tip 16.3 mm from its place in the single Panda's hand. It was not a slide along the handle: on one stroke the hand came 5.6 mm under its streamed height, the blade pressed on the bowl (45.5 N for 220 ms with the mixture), and the handle turned about the line through the two pads; the tip was read with the blade still on the floor and its own give in it. `Operator.follow` therefore waits for the arm (`keep_up`, `KEEP_UP_RAD` = 0.02 rad, on by default for a Panda, `--keep-up` for an FR3): with the mixture over 20 strokes the hardest knock was 28.0 N and the longest 18 ms, and the handle stood within 0.28 degrees of its place at the end of every stroke. The Panda's URDF carries its own finger collision mesh beside the Franka finger boxes the scene adds, 0.09 mm behind the pad's face. Left on, the mesh takes half of every pinch and the two surfaces' frictions pull against each other with 10 to 11 N while a 0.09 kg tool hangs still. `scene._fingertips` switches that mesh's collision off, so a Panda finger has the four boxes an FR3 finger has. In the robot runs of section 11.5, with the food the right Panda alone folded the muffin batter and blended the sauce, and the Panda pair cooked the custard, with no protective stop; at one of three marks of the muffin run the spatula's tip read 20.9 mm from its place in the hand (2.8 and 1.7 mm at the others, read with the blade at the bowl).

A sliding body (`world._contact_rule`, `params.SLIDE_REFERENCE`): a friction row's reference speed is capped at what Coulomb friction can actually decelerate, rather than MuJoCo's own reference (which threw a body sliding faster than 6 mm/s off the surface); pads keep their own rule (section 11.8).

GAR (`gym_anything_robotics/backends/newton_ingredient_mixing.py`). Actions: `named_joint_target`, `gripper`, `cartesian_pose`, `cartesian_delta`, `keyboard`, and the raw operation `set_load`. Observations: RGB from up to four cameras; joint positions, velocities and measured torques; finger widths; `robot_status`; on the hob tasks `instruments.thermometer_`

c, the probe thermometer's own reading (20 C while it stands in the crock). Privileged: state, geometry, evaluation, checkpoints, per-tick trace, variation selection, diagnostic cameras.

A command sent again with its key gets its stored reply. A request is checked against the command floor when it is decoded (_precheck), and the gateway looks its Idempotency-Key up after decoding it. A refusal raised in the decoding would answer a command that had been carried out, sent again after the floor had risen, with "rejected before execution". So _precheck only notes a refusal, and act_policy raises it when the request's first command is about to run, after the key has been looked up and before anything moves. On a live sauce-blend session: a 64 ms hold committed, the render worker ended, the floor at 103 ms, the same body and key sent again: 200, committed, the same request id, no simulated time added. A unit test raises the floor between the two sends. A refused request leaves one failed row in the session's history, as a pose out of reach does.

Long commands. gar action waits 120 s for a reply unless given --timeout; a command is answered when its physics and its observation are done, and nothing bounds its length (a keyboard chord may last 10 s, a joint target any number of ticks). The session says what to expect: hardware.budget carries the wall seconds expected per simulated second, the result for 1 and 10 simulated seconds and the longest command the default wait covers, and hardware.long_commands, the policy's /v1/help and the owner's capability note say in words that such a command needs --timeout and that a reply that was not waited for is asked for again with the same key, not sent as a new command. On a live session a 2 s hold sent with --timeout 8 timed out in the CLI after 8.25 s, and the same line with the same key and a longer wait came back committed with the simulated time advanced by 2.000 s once.

The command floor: a session starts at 21 to 154 ms depending on its cameras, then follows the dearest of the last three commands, never below the published floor and at most 2 s. What it costs through the gateway, and with how loaded a machine, is in section 11.8 (193 policy commands). The render worker starts beside a command's physics, is kept for idle_release_s (default 900 s), and is replaced if its process has died; renderer options (width, height, samples, idle_release_s) are refused at launch if a key is unknown, a value out of range, or the floor they imply exceeds the 2 s cap.

Policy camera	Where it is	What it shows
station_rgb	wall bracket behind the vessel, 0.69 m above the counter, 84 degrees	the vessel's contents, both hands, the crock, the mixer or the knob
overhead_rgb	under the shelf, straight down, 40 degrees	the inside of the vessel
left_wrist_rgb, right_wrist_rgb	D405 envelope on each hand	what the hand is over or holds



Figure 6. What a policy is given. Top: station_rgb and overhead_rgb at the start of muffin-fold, station_rgb at the start of stir-sauce. Bottom: the two wrist cameras at the ready pose, and overhead_rgb over the pan.

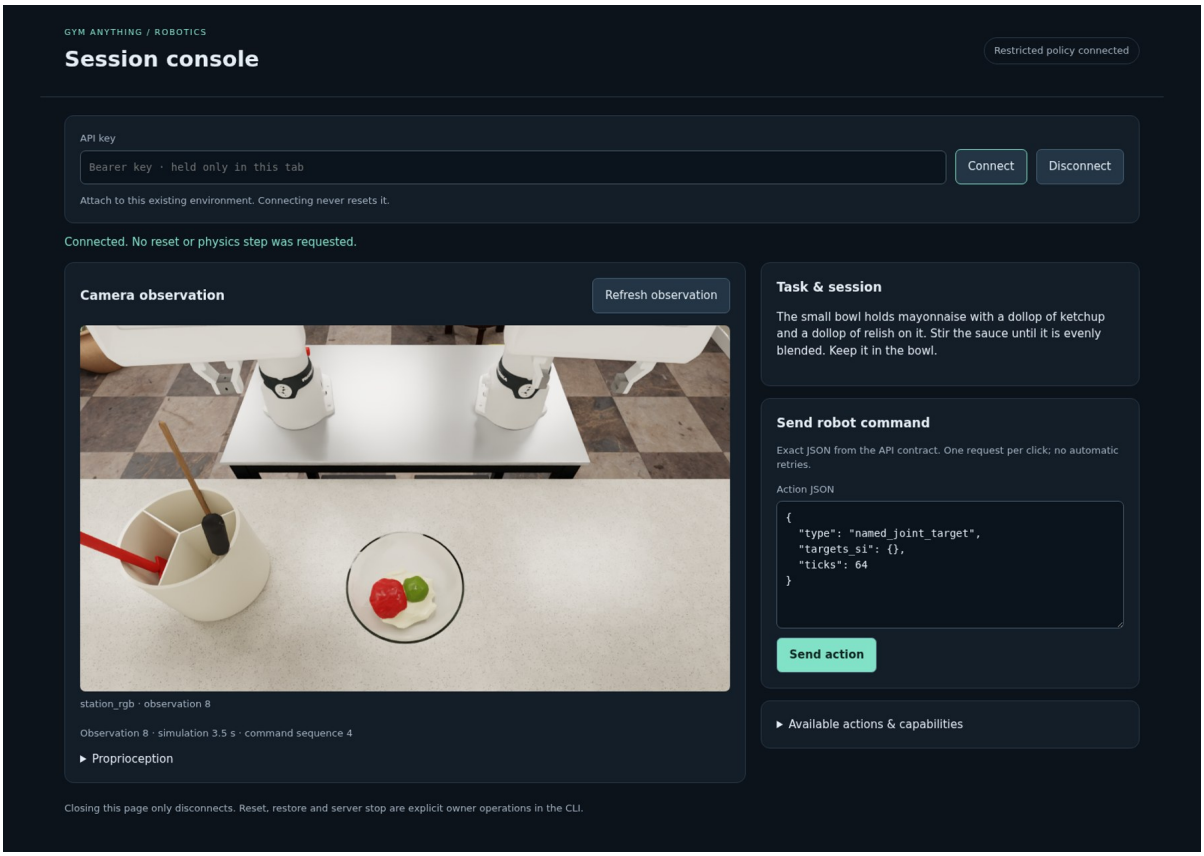


Figure 7. The shared dashboard on a live `sauce-blend` session, opened in a browser with the policy's key (section 11.8): the task, the cameras a policy gets, the action form. The key field is cleared after connecting; no key is in the picture.

10 / Tasks and what the evaluator measures

Eight tasks in two presets. Each starts with the ingredients in the vessel as a cook leaves them and asks for an outcome.

The eight tasks, their recipes and limits are in `task.py` (`TASKS`). A start state is put in as a person would leave it (`task._Fill`): a heap of powder with its own slope, a well in it, a liquid poured round it or into it, spoonfuls dropped on a paste. Each ingredient takes exactly its recipe's volume of the points' lattice: the points that lie lowest in a potential that describes the shape (height plus a slope times the distance from the heap's axis, and a few seeded waves so that no outline is a circle and no top a plane); a seed turns the layout and moves it by up to 12 mm. The hand-over waits until the solver holds what was put there: the points' 99th-percentile speed under 1 mm/s at three checks in a row, for at most 12 s, after which the state is handed over as it is and its record says so. All eight are handed over at rest, after 0.7 s (`pudding-whisk`, `stir-sauce`, `reheat-porridge`) to 3.9 s (`cake-mixer`), the 99th-percentile speed 0.12 to 0.89 mm/s, the 99.9th 0.18 to 2.29 mm/s and the fastest point 0.35 to 4.35 mm/s. "At rest" means the 99 % only: the record also gives the 99.9th percentile and the fastest point, because liquid soaking into powder and a spoonful spreading go on for minutes.

Task	In the vessel at the start
`dry-whisk`	a heap of flour, a small heap of sugar against one side of it, the baking powder and salt against another
`muffin-fold`	the dry mix as a ring with a well in the middle; milk, egg and oil in the well
`pancake-whisk`	the dry mix heaped in the middle; milk, egg and oil poured round it
`pudding-whisk`	the pudding powder heaped; cold milk poured round it
`cake-mixer`	the cake mix heaped; water and oil poured round it; three eggs dropped on top
`sauce-blend`	mayonnaise spread on the bowl's floor; a spoonful of ketchup and one of relish on it
`stir-sauce`	cold milk with the starch and the sugar stirred in, level in the pan; the hob on at 1000 W (the knob's twelfth position, the manual's level 6.0); a probe thermometer in the crock
`reheat-porridge`	cold set porridge spread in the pan, its top uneven; the hob on at 580 W (the seventh position, level 3.5); a probe thermometer in the crock



Figure 8. The eight start states as built and settled, from the close camera. Top: `dry-whisk`, `muffin-fold`, `pancake-whisk`, `pudding-whisk`. Bottom: `cake-mixer`, `sauce-blend`, `stir-sauce`, `reheat-porridge`.

The evaluator (`task.evaluate`) reads the points and the rigid state; a policy cannot read either.

- *Where the mass is*: inside the vessel (below the rim, within the inner surface, in the vessel's frame), on the counter, on the floor, airborne, carried as a film on wires or beater strips, elsewhere (on a tool, on the rim). *Contained*: counter, floor and airborne at most 1 % of the batch; for `stir-sauce` also what lies outside the pan below its rim (on the hob's glass beside it, down its outside), which stays active mixture inside the grid region and so is counted here by name. The total is compared with the batch at every evaluation (mass balance). What is outside the vessel is also reported by how it left it: over the rim, below the rim, or with no record.
- *Uniform*: how unevenly each ingredient the task puts in is spread through what is in the vessel (`task.uniformity`). The material in the vessel is cut into n portions of EQUAL MASS, n the batch's volume over one portion's (at least 8): the set of points is halved by mass across its longest extent, and each part again, until there are n (`task.equal_portions`). Each ingredient's mass fraction is read in each portion and its coefficient of variation taken over the portions. That is done in four frames turned about the vessel's axis by 22.5 degrees and the four are pooled (root mean square), so that the reading does not hang on where the first cut falls; the lowest and highest of the four are reported beside it. The least evenly spread ingredient is judged. At least half of the batch must be in the vessel, or the criterion fails whatever the reading. Every point is in exactly one portion in every state, so nothing comes into the reading or leaves it as the

material moves. (Portions fixed as cubes of the grid would come into the count and leave it as a whisk heaps the powder, and the reading of the dry mix would jump by up to 0.54 between marks two seconds apart.) How much the reading of one state moves when nothing about its mixing has changed: the dry mix's start state read five times with every point shaken by a fifth of the point spacing reads 2.0439 to 2.0453 against 2.0458 unshaken. An even mix of a tracer that is every twenty-fifth point reads its counting noise, 0.065 over twelfths (unit test `Unevenness`).

- *No dry powder*: mass of the solid still dry over all of the solid. *Smooth*: mass in lumps over the solid. (`stir-sauce` does not judge lumps: once its starch is cooked no state its start allows has cooked starch in 6 mm cells at more than twice the batch's mean fraction, so such a criterion could not fail; see the comment above `task.TASKS`.)
- *Not over-mixed*: the work the batter has taken per kilogram where its wetted flour is (the points' plastic and viscous work, averaged over the wetted flour's mass), against 25 reference fold strokes of 0.33 J/kg each (8.25 J/kg; the stroke is measured in section 11.2).
- *Cooked*: the gelatinised share of the starch in the pan, and the highest mean temperature the sauce reached: the hob's record of the mass-weighted mean of the sauce inside the pan, taken after every step and kept in every checkpoint (`heat.Hob.food_peak_c`), or the mean in the pan now if that is higher. The share is judged as it is reported, to 1e-4. (The highest mean is judged, not the mean at the moment of judging, because a cooked sauce cools once the hob is off and the verdict would hang on how long after that it was read.) *Hot through*: the coldest 60 ml portion's temperature at the moment of judging and the spread between the coldest and the hottest portion. Both are judged as they are reported, to a hundredth of a degree. (A rule that judged them on the highest temperature each point had ever had passed a porridge whose coldest portion was at 66.6 C: it follows the pan, not the food.) *Not scorched*: the area of burnt-on patches.
- *Vessel in place* (within 50 mm, upright, not lifted), *appliance off* (the hob: knob at 0; the mixer: switch at OFF, beaters under 5 rpm, the mixer more than 40 mm outside the bowl's rim, at rest and not held), *tool out* (no point along a tool's working half inside the vessel).

Task	Criteria and limits (`task.TASKS`)
`dry-whisk`	baking powder and salt, and sugar, each uniform over twelfths of the batch, CV at most 0.10; contained; bowl in place
`muffin-fold`	flour uniform over twelfths, CV at most 0.10; dry flour at most 2 %; mixing work at most 25 reference strokes; contained; bowl in place
`pancake-whisk`	flour uniform over 60 ml portions, CV at most 0.10; dry flour at most 2 %; mixing work at most 25 reference strokes; contained; bowl in place. Lumps are not judged
`pudding-whisk`	mix uniform over 120 ml portions, CV at most 0.10; dry mix at most 2 %; lumps at most 2 %; contained; bowl in place; whisk out of the bowl
`cake-mixer`	mix uniform over 60 ml portions, CV at most 0.10; dry mix at most 2 %; lumps at most 2 %; contained; bowl in place; mixer off and set down
`sauce-blend`	mayonnaise, ketchup and relish each uniform over 5 ml portions, CV at most 0.10; contained; bowl in place
`stir-sauce`	at least 95 % of the starch in the pan cooked and the sauce having reached 88 C or more; burnt on at most 2 cm2; contained, counting sauce outside the pan below its rim; the pan upright at the hob's height with its whole base (its base disc, corners at 66.3 mm) on the hob's glass; knob at 0; the sauce at the boil (mean at 96 C or more) with the hob on for at most 180 s in all. Lumps are not judged
`reheat-porridge`	every 60 ml portion at 74 C or more and none more than 15 K hotter than the coldest; burnt on at most 2 cm2; contained; pan in place; knob at 0

Where the limits come from (the comment above `task.TASKS`):

- *Unevenness 0.10.* The feed industry's mixer test calls a tracer's coefficient of variation under 10 % "excellent" (Herrman and Behnke 1994, as quoted in Kansas State University bulletin MF3393, opened); the pharmacopoeia's content uniformity is 6 % (opened). No baking standard for leavening or salt was found.
- *25 strokes' worth of work.* "15 to 20 light strokes" is the muffin method's direction (Quaker Oats, Baking 101, opened); 25 is that with a margin, authored. The reference is one of the method's own strokes, 0.33 J/kg.
- *Dry flour 2 %.* "Just moistened"; the 2 % is authored. The wetting constant is calibrated so that the method's strokes reach it (section 11.2).
- *`pancake-whisk`: no dry flour and not over-mixed, lumps left alone.* The recipe the batch is measured from says "Stir into the milk/egg. Don't beat; some lumps are OK" (King Arthur, opened).
- *74 C for the porridge.* 165 F: "The FDA Food Code requires that all leftover foods ... be reheated to 165 F for 15 seconds within 2 hours" (Institute of Child Nutrition, Reheating Foods fact sheet; opened), and "Leftovers must be

reheated to at least 74 C (165 F)" (Government of Nunavut, Leftover Food Factsheet; opened). The United Kingdom's table has 70 C held for two minutes or 75 C for 30 s. The spread of 15 K is authored.

- *88 C for the sauce.* 190 F, the lower end of a maker's range for a milk sauce thickened with starch: "The pastry cream should reach 190 F to 200 F on a digital thermometer", after it has thickened and bubbled and been whisked "for 1 additional minute" (King Arthur Baking, Pastry Cream; opened). The minute at heat is in the other half of the criterion, 95 % of the starch cooked by the gelatinisation kinetics. 88 C is below this hob's simmer (no vapour leaves as bubbles below a mean of 96 C, `params.HOB` simmer_c): the instruction names the simmer as the cook's sign of done, and the verifier asks only for the maker's lower figure.
- *Lumps 2 %, 95 % cooked, 1 % spilt, 50 mm:* authored tolerances with no outside figure. None was found for a batter's lumps.
- *2 cm2 burnt on:* authored. The bottom's patches grow outward (0.24 cm2 at the centre to 3.5 cm2 at the rim), so one burnt patch passes in the inner four rings and fails in the outer four, where the heating ring puts 80 % of the hob's power.

A task succeeds when every criterion holds; the score is the share that hold (two of three for an untouched `sauce-blend`), except `stir-sauce`'s: 1 on success, else 0.9 times its progress, the gelatinised share of the least cooked of three equal-mass layers of the sauce in the pan, by height, over 0.95 (at most 1), times the share of the batch still in the pan times the share of the bottom not burnt on times a closing factor (0.75, + 0.10 with the pan on the hob, + 0.15 with the hob off over a cooked sauce; the weights and the linear burn factor authored), read from the final state (`task.cook_progress`, `task.layer_shares`). The layers keep a bottom layer cooked on an unstirred pan under cool milk from counting as the sauce thickening. The share of criteria would give an untouched pan 4/6 and a knob turned to 0 with nothing cooked 5/6, the same as a whole bottom burnt on. Section 11.8 tests each criterion on a pair of written states, one each side of its limit. No instruction tells the motion; the porridge's instruction says that a probe thermometer stands in the crock, and the sauce's names the simmer as the sign of done and the thermometer as a thing in the scene.

11 / Experiments and results

Each subsection: what was run, what it shows, the record. ASSISTED runs move the world with something no policy has (a phantom carriage, force or placement); ROBOT ACTIONS runs use only the public action interface's controls.

11.1 The solver against closed-form results

The cup gate, the regional rest measurements and the wall-thrust trials were run on the solver as it is. The wall calibration, the square-box rest, the tipped vessels and the jerked bowl were run on an earlier version of the solver, as stated below.

A cup of inner radius 40 mm with a concentric rod of radius 20 mm that ends 10 mm above the floor holds 80 mm of a material whose law is set directly. The rod turns; the torque on the rod and on the cup are compared with laminar Couette flow between the cylinders plus the torsional flow under the rod's end. The wall takes tangential velocity at the rate $c_w \nu / \Delta x^2$ (section 4), with c_w = 8 calibrated on the 3 Pa s row (on the solver as it is, that row reads 0.99); that row is the fit and is not evidence of anything.

Torque (cases `torque_dx2.5`, `torque_dx2`, `torque_dx4`, `torque_dx2.5_reflected`): measured over laminar (Figure 9).

Material	K (Pa s^n), n	Rod speed	2.5 mm cells	2.0 mm	4.0 mm	2.5 mm, mirrored walls
Newtonian	100, 1	5 rpm	1.00	1.01	0.97	0.92
Newtonian	30, 1	20 rpm	0.99	1.02	0.94	0.92
Newtonian	10, 1	60 rpm	0.96	1.00	0.93	0.95
Newtonian	10, 1	120 rpm	0.92	0.96	0.91	0.87
Newtonian	3, 1	60 rpm	0.99	1.02	1.01	0.98
Newtonian	1, 1	60 rpm	1.06	1.07	1.17	1.06
Newtonian	0.2, 1	60 rpm	1.35	1.27	1.83	1.36
Newtonian	0.01, 1	60 rpm	4.20	3.88	4.68	4.05
Muffin batter	23, 0.56	20 rpm	1.03	1.04	1.04	0.98
Muffin batter	23, 0.56	60 rpm	1.02	1.04	1.05	0.93
Spelt pancake batter	6.9, 0.325	60 rpm	1.22	1.20	1.43	1.12
Pancake batter	0.754, 0.676	60 rpm	1.29	1.24	1.67	1.28
Pancake batter	0.754, 0.676	180 rpm	1.35	1.28	1.83	1.23

From 3 to 100 Pa s the torque is within 4 % of the laminar result at 60 rpm and below, and the muffin batter's power law within 3 %; at 120 rpm the 10 Pa s row is 8 % low. Thinner, the torque is over: 6 % at 1 Pa s, 35 % at 0.2 Pa s, 29 to 35 % for the wheat pancake batter, 22 % for the spelt. At 0.01 Pa s, five times milk's viscosity, it is 3.9 to 4.7 times the laminar result at the three cell sizes: such a liquid is not resolved (section 12). The torque on the cup equals that on the rod within 0.3 % in every row down to 0.2 Pa s (`balance`); at 0.01 Pa s they differ by 15 %.

Volume (`volume_60s`): a 3 Pa s material stirred for 60 s at 60 rpm: its centre of mass falls from 50.08 to 49.63 mm (0.9 %) and not one of 161,120 points is outside the cup at any mark. *Slumps* (`slump_50/100/200`): a column 80 mm high of a material with a yield stress of 50, 100 and 200 Pa slumps by 0.62, 0.50 and 0.24 of its height, where Pashias and others' cylinder slump gives 0.63, 0.43 and 0.18. *Rest* (`rest_30s`, `rest_muffin_30s`, `rest_thin_30s`): a material with a yield stress of 60 Pa, and the muffin batter, left alone in the cup for 30 s, move 6.5 and 1.7 micrometres at the median, 21 and 11 at the 99th percentile, and their fastest points are slower than 0.3 micrometres a second at the end. A thin liquid (1 Pa yield stress, the rest stress every wet state has) does not come to rest: 1.4 mm at the median in 30 s, 13.7 mm at the 99th percentile, its 99th-percentile speed about 1 mm/s throughout.

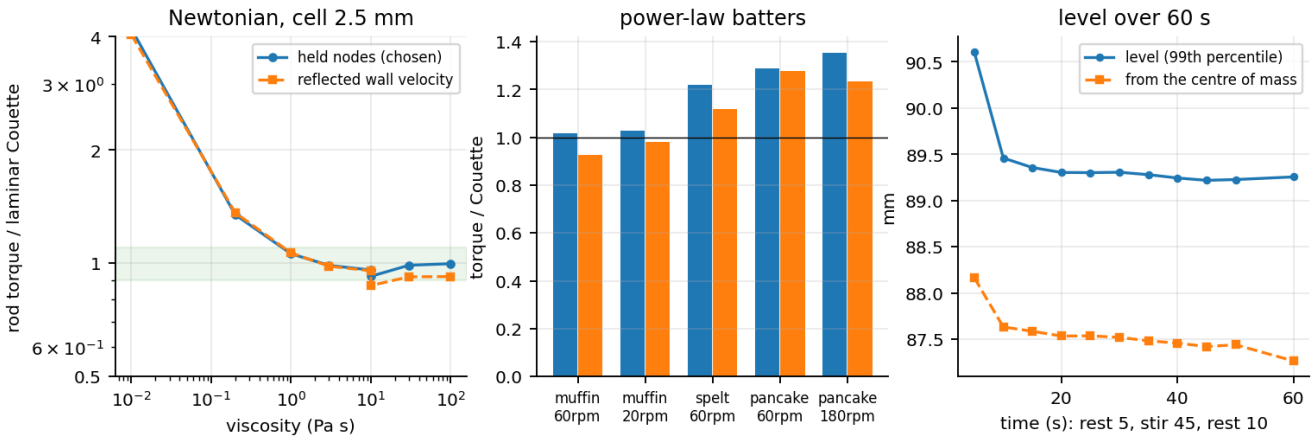


Figure 9. Stirred-cup torque against the laminar reference, and the cup's level during the volume check. Thin liquids give more torque than the laminar result.

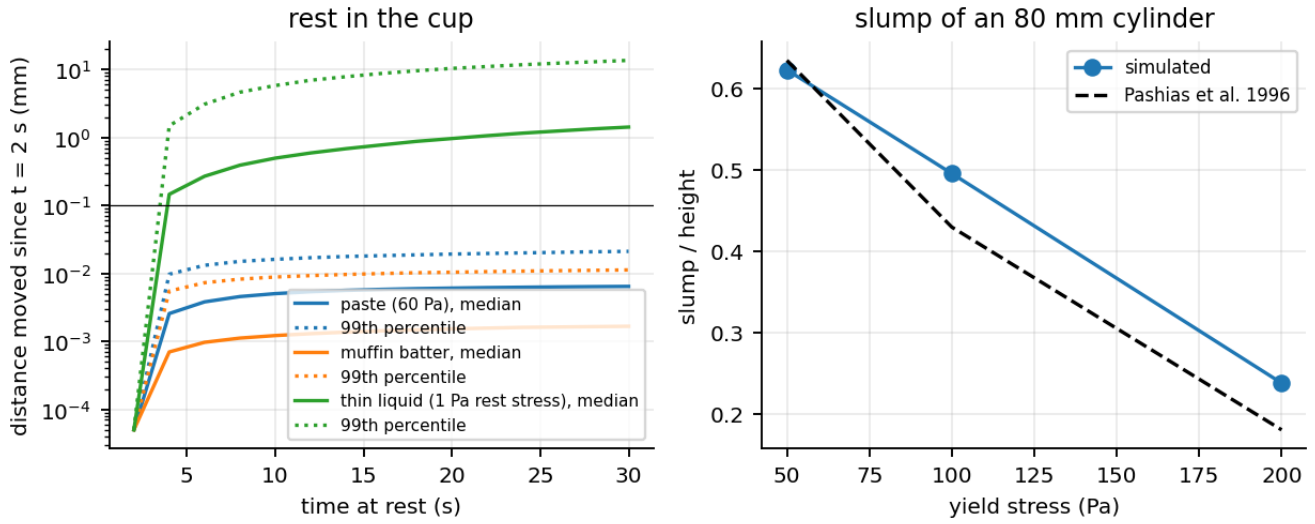


Figure 10. Rest and yield-stress slump in the cup. The thin liquid keeps moving while the paste and muffin batter settle.

Where the thin liquid creeps (the same liquid, left alone 12 s, read by region). Median 0.59 mm; in the corner of floor and wall 5.7 mm, moving outward and up (234 and 219 micrometres a second); on the floor 1.18 mm; next to the outer wall 0.83 mm; at the free surface 0.33 mm. Holding every wet wall node gives 0.27 mm (0.05 in the corner) and costs the torque rows; without the look-ahead at corners 1.05 mm, without the keep-out rule 0.71 mm. The creep comes from a force the solver leaves on the nodes next to a wall: the pressure's thrust on a wall node, gathered from points on one side of it only, has a part along the wall. Taking that part out in the grid update (`k_grid`, `wall_thrust`; off in the solver as it ships, `wall_thrust` 0) was tried in two forms (Table 11.1a).

Case	Median	Corner	Floor	Outer wall	Free surface	Rod and cup torques apart, 3 / 1 / 0.2 Pa s
as built	0.59 mm	5.7 mm	1.18 mm	0.83 mm	0.33 mm	0.05 / 0.08 / 0.3 %
the wall-parallel thrust removed	0.23	0.02	0.05	0.09	3.18	6 / 43 / 130 %
the same, not next to a free surface	0.87	8.2	3.2	1.26	0.48	57 / 101 / 118 %

Removed everywhere, the thrust stops the creep at the walls and moves the free surface ten times as far, and the torque on the cup no longer equals the torque on the rod, which in a steadily stirred cup it must: the change itself puts torque into the liquid or takes it out. Kept away from free surfaces, it makes every region worse. Neither cures it; the creep is not solved (section 12).

Run on an earlier version of the solver (nothing in these runs has a wire, a cut or a hob): a thin liquid left 120 s in a square box (0.12 mm at the median, the farthest point 5.4 mm); the tipped saucepan, bowl and sauce bowl (a thin sauce starts to

pour at 44 degrees where its geometry says 39.3 and pours too slowly); the jerked bowl of milk (it sloshes at the right period, 0.79 to 0.81 s against 0.82 s by linear theory, and loses its swing at 5.4 to 13.4 % of critical damping where water has about half a per cent).

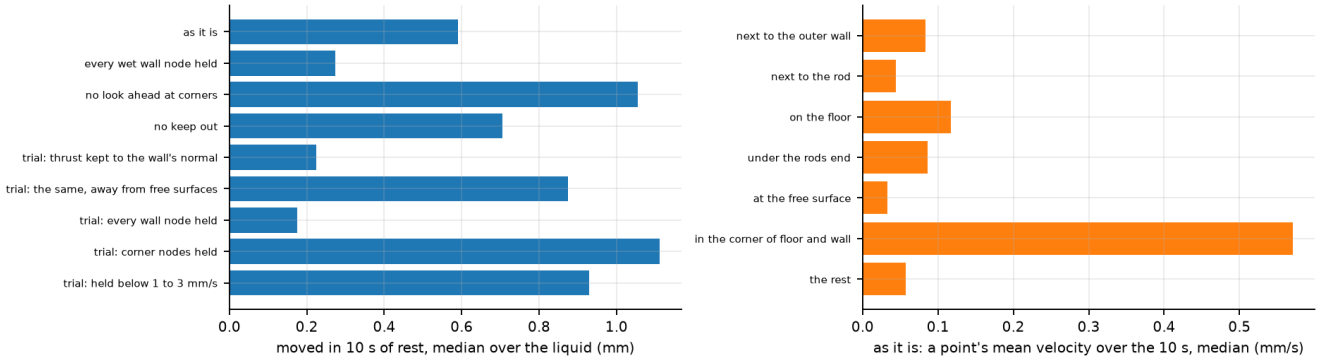


Figure 11. Resting displacement by region and the rejected wall-thrust changes. Reducing motion at the wall did not preserve the free surface or torque balance.

11.2 The material's laws on benches

No robot and no room: a box or the cup of section 11.1, the recipe's own material.

The wetting constant is set on the muffin method's own direction, "mix just until the dry ingredients are moistened, 15 to 20 light strokes" (Quaker Oats, Baking 101), read as 2 % of the flour still dry: 2e-6 m2/s reached it at 17.6 strokes of 160 mm across the bowl, and 5e-7 at about 31. It separates those two and does not separate 2e-6 from 4e-6 or 8e-6; the lower was kept. It is a calibration on one outcome with one path, not a measurement of how fast flour takes up milk. The same strokes after the 13 s wait leave 4.5 % of the flour dry at 30 s and 2.35 % at 40 s with 9.4 of the 8.25 J/kg allowed spent: the batter is over-mixed before it is moistened with strokes that do not reach the bowl's edge (section 11.6 has the strokes that do). The reference stroke of the over-mixing limit, 0.33 J/kg, is the work a calibration stroke put into the wetted flour after 16 strokes.

Batters in the cup (E7): measured torque over the Couette result for the recipe's own law, at 20, 60 and 180 rpm.

Batter	Law at 20 C	Apparent viscosity at 10, 60, 200 1/s	Torque over theory
Pancake, wheat	K 1.16, n 0.66	0.53, 0.29, 0.19 Pa s	1.17, 1.23, 1.29
Pancake, spelt	K 10.8, n 0.30	2.16, 0.62, 0.27 Pa s	1.13, 1.20, 1.29
Muffin	K 24.3, n 0.53, 24 Pa	10.6, 3.96, 2.14 Pa s	1.02, 1.01, 1.00

The muffin batter is resolved; the pancake batters read a fifth to a third stiff, as the thin rows of section 11.1 do.

A column of flour let go (E5; Drucker-Prager, 37 degrees, cohesion 80 Pa). A column 30 mm across and 180 mm high, above the critical height of 102.5 mm the cohesion allows, collapses to 28 mm high and runs out to 135 mm in 1 s and stays (Lube and others' dry sand at the same aspect: 36 and 148 mm); one 90 mm high, under the critical height, slumps only to 51 mm high and 52 mm out, where cohesionless sand runs to 113 mm: the cohesion holds it.

Liquid into a bed of flour (E6): after 60 s at the model's wetting constant the front is 7.4 mm into the bed and 83 % of the bed's flour is still dry; at three times the constant and at a third of it the front reaches the same 7.4 mm by 30 s (83 and 85 % dry): the front's depth does not follow the constant; why was not traced.

Mixing below a point's spacing (E10): a tracer stripe in a paste at rest, its width fitted to a diffusion law. Set to 1e-6 and 4e-7 m2/s the closure gives back 0.99 and 0.96 of the set value; at a molecule's 5.2e-10 m2/s it gives 0.38 to 0.44 of it (the stripe grows to 0.17 mm in 30 s, about an eighth of a point's spacing): below the points' scale the closure is a floor, not a measurement.

Lumps (E11; strength 20 Pa, erosion 0.05): left at rest or stirred by the rod at 60 rpm, every lump of a pancake batter is still there after 8 s; drawn through by the whisk's wires (45 Pa at the wire), 78 % is left; with the strength a third, 46 %; three times, all. In the muffin batter the rod leaves 47 % and the wires 42 %. Neither the strength nor the erosion rate has a measured basis (section 12).

Time laws (E13): the pudding's mix, wetted, thickens from 0.03 to 2.4 Pa s at 50 1/s over its setting time of 100 s; flour hydration runs on a 300 s time constant; the batter's stiffening with work runs on a scale of 78 J/kg (about 240 reference strokes).

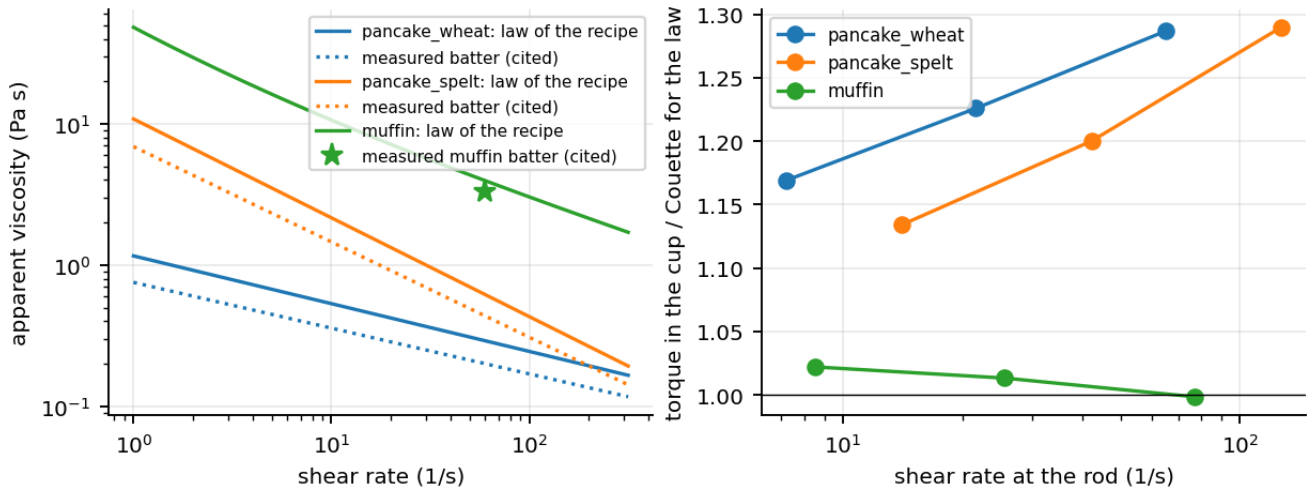


Figure 12. The batter laws and their measured torques in the cup.

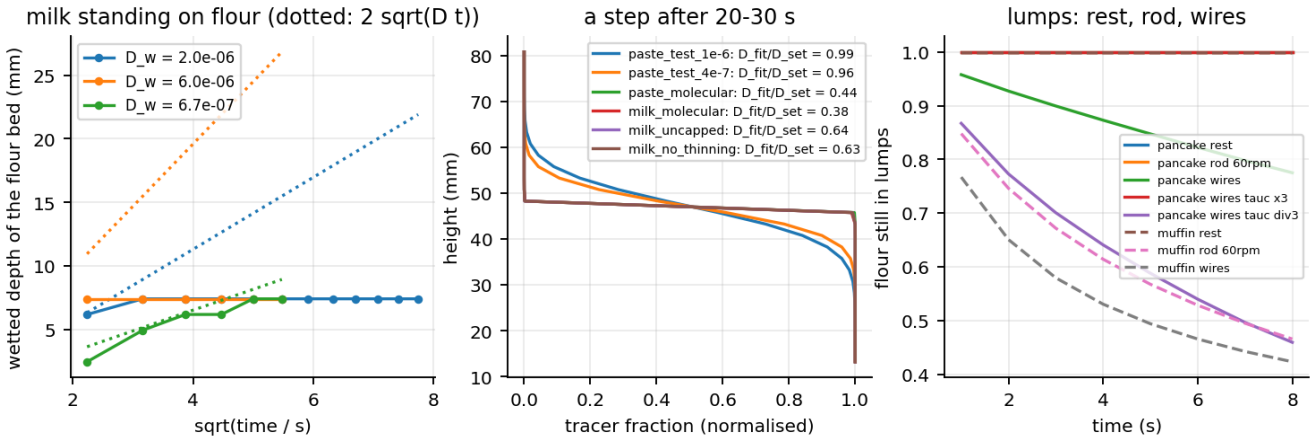


Figure 13. Material benches: wetting, composition exchange and lump erosion. These measurements expose the stated limits of the closures.

11.3 Tools and appliances

One wire, resolved (Figure 14). A whisk's wire is 1.4 mm across and the tasks' cell is 2.5 mm: the solver cannot see the flow round it. Here the same solver resolves the wire: a solid capsule on cells of 0.20 to 0.40 mm, drawn through a slab of the same flour (Drucker-Prager, 37 degrees, cohesion 64 Pa, 550 kg/m3) in a box whose two side walls are frictionless, so that the slab is in plane strain. The bench is first held to an outside answer. In a purely cohesive material (200 Pa, no friction), drawn slowly 10 mm deep, the resolved wire meets 8.71 and 8.54 times strength times diameter on 0.35 and 0.25 mm cells, where the limit load of a cylinder is 9.14 (smooth) to 11.94 (rough) (Randolph and Houlsby 1984): within 5 to 30 % of it, and below it, as a slightly coarse resolution of a smooth wire should be.

In the flour: an upright wire drawn sideways at 0.25 m/s leaves the powder in its own track 3.1 to 3.5 mm farther along, the powder half a diameter beside it 1.8 mm, one diameter away 1.1 mm, two away 0.46 mm (Figure 14, left); summed over the points, 10.4 and 10.5 mm2 of powder carried per length of wire on 0.40 and 0.30 mm cells, the two cell sizes within 1 %. Half of the powder that was in the wire's own track is left there (the cut). At 0.60 m/s 14.1 mm2. A wire lying 10 mm under the surface carries the powder above it farther, 22.6 to 23.2 mm2 (24.3 at 0.60 m/s). Two wires side by side: three diameters apart the powder between them goes along as a plug, 21.1 mm2 a wire; five apart 11.4; eight apart the slab's side walls are 4.4 mm from each wire and the case is not used.

The solver's own wire under the rule, a chain of its markers thinner than a cell, in the same benches on the tasks' 2.5 mm cells and on cells half that size (Table 11.3a). The rule (section 4: a dry point near a wire goes the wire's way at no less than 0.5 exp(-(r - d/2)/(0.7 d)) of the wire's speed) was fitted to the resolved upright profile at 0.25 m/s; it acts on the points, and in that bench its result is the same on both cells within 4 %, four fifths of the resolved wire's. The simpler rule (the wire's drag, and a point inside the wire's cylinder set on its surface; no carry, no cut), switched on in the same bench and read the same way, carries 0.27 and 3.5 mm2 upright and 2.7 and 6.2 mm2 lying 10 mm deep on the two cells: a factor of 13 and of 2.3 between them. Where the bench departs from the one it was fitted on, the rule falls short of the resolved wire and the grid's

own share comes back in: drawn at 0.60 m/s, or lying under the powder, the solver's wire carries less than the resolved one and more on the finer cell. Two wires three diameters apart are carried as a plug on 2.5 mm cells, where a single cell spans the gap (19.2 mm2 a wire against the resolved 21.1), and not on 1.25 mm cells (10.1): there the result still depends on the cell, by a factor of 1.9. The wire's drag in the flour is below the resolved wire's on both cells and grows as the cell shrinks.

Bench (flour, 0.25 m/s unless said)	Resolved, 0.25 to 0.40 mm cells	Solver, 2.5 mm cells	Solver, 1.25 mm cells
upright, carried per length (mm2)	10.4, 10.5	8.4	8.1
the same, drag and keep-out only (mm2)		0.27	3.5
upright at 0.60 m/s (mm2)	14.1	8.4	12.8
lying 10 mm deep (mm2)	22.6, 23.2	8.4	10.3
the same, drag and keep-out only (mm2)		2.7	6.2
two wires 3 d apart, per wire (mm2); powder midway (mm)	21.1; 7.1	19.2; 3.6	10.1; 1.9
two wires 5 d apart, per wire (mm2)	11.4	9.3	12.8
drag on the upright wire (N/m)	3.7 to 3.8	1.8	2.9

The 8 d pair is left out: in the resolved slab its side walls stand 4.4 mm from each wire.

The liquid that follows a cut. The same resolved wire drawn at 0.25 m/s from a pool of thin liquid (0.01 Pa s) into a bed of the flour standing beside it, read 0.4 s after the pass. 71.0 and 71.2 mm3 of liquid follow the wire into the bed on 0.40 and 0.30 mm cells, for 11.9 mm of wire in the bed: 6.0 mm2 per length, 3.05 times the wire's diameter squared, all of it within two diameters of the cut. A cut twice as long takes 68.4 mm3 (2.94): the liquid is a dose taken in where the cut meets the pool, not a share of the cut's length. At 0.60 m/s 3.15. Into a stiff paste (the cohesive material, 200 Pa) 42.0 mm3, 1.80. From a liquid of 1 Pa s 91.5 mm3, 3.93. A round wire 4 mm across, the width of a beater's strip, at 0.60 m/s: 2.63 times its own diameter squared (no flat strip was resolved). Where the liquid lies along the cut is not settled: within 6 mm of the bed's face on 0.40 mm cells, along all 15 mm on 0.30 mm cells. A heap free to give way takes more than a bed: a heap 12 mm wide standing in the pool takes liquid into 10 of the cut's 15 stretches, 0.80 diameters per unit of cut area, and the one wire pushes 13 % of its powder out of its footprint; 17 mm wide, 7 of 15 stretches, 0.43, 2.7 %. The rule (section 4) takes 3.0 wire diameters squared from a thin liquid and 1.76 into a paste, everywhere: it is the bed's figure, it understates a small or loose heap and a liquid of 1 Pa s (by about a quarter), and the beater's strip is held to a round wire of its width.

The cage against a blade on the task's own heap is in section 11.6.

The tools on benches (ASSISTED, a phantom carriage carries the tool). A spatula's fold stroke of 170 mm at 0.20 m/s through the muffin batter puts 0.34 to 0.38 J/kg into it a stroke, peaks at 5.8 to 7.2 N, and moves the bowl standing on the counter 0.34 to 0.36 mm a stroke (5.5 mm over the bench); standing on the silicone mat it does not move. The spoon drawn broadside through mayonnaise meets 0.62 N at 0.05 m/s and 0.89 N at 0.20 m/s, against an order of magnitude of 0.57 and 0.78 N from six times the yield stress times the blade's area. The whisk drawn through a mixed pancake batter meets 0.054 N at 0.15 m/s and 0.107 N at 0.35 m/s, against 0.059 and 0.146 N from its wire law in still batter.

The hand mixer (ASSISTED: a phantom carriage holds the mixer). Its slide switch moves through its five settings at 3.06 to 3.15 N of push and rests in each detent; the motor runs free at the measured 288 rpm on its first setting and 1107 on its fifth, on a straight torque-speed line to its stall (216, 144, 72 rpm at a quarter, a half and three quarters of it). In the unmixed cake batter on its first two settings the beaters turn at 281 to 288 and 481 to 488 rpm against 288 and 493 free; in 9 s at those settings the dry share of the mix falls from 0.80 to 0.54, and at most 0.02 % of it is counted outside the bowl. Switched on at its fifth setting at once, against the maker's direction, it throws batter as high as 128 mm over the floor, 14 mm over the rim, and none over it (section 12, limit 9). With the wire's means divided by the mass gathered with them (section 4), 200 random placements of fast drops beside the strips read nothing that is not finite, and the largest viscosity read is 3.0 Pa s, the liquid's own.

The knob (ASSISTED, a phantom torque on the knob's joint): with the detent at 0.035 N m, a torque of half or 0.85 of it leans the knob 3.75 and 6.4 degrees in its detent; 1.15 times it turns the knob through every detent to its stop, off or the twentieth level. **The spoon's bowl**: the spoon rests on its back, hollow up, and a pea and a cube dropped or shoved stay in its hollow (shoved across a flat slab of the same outline, the pea goes off it).

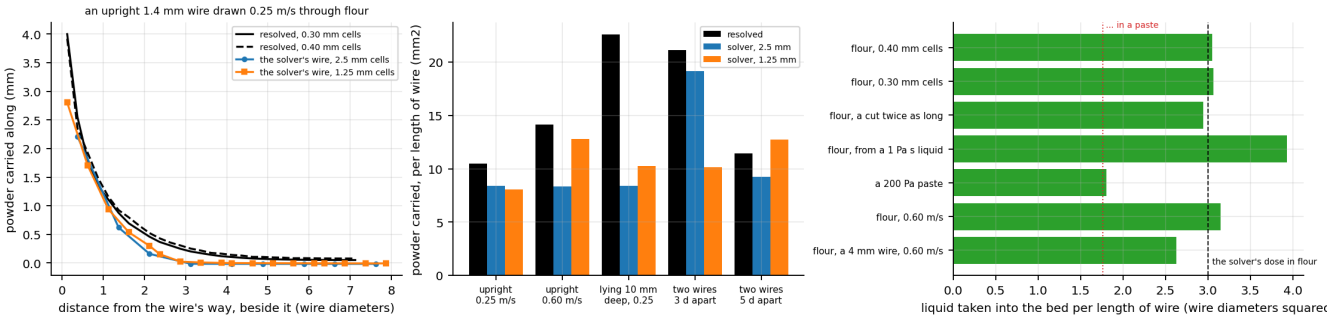


Figure 14. A resolved wire and the sub-cell rule: displacement profiles, carried powder and liquid entering a cut. The agreement is strongest in the fitted upright case.

11.4 Heat on the hob

A sauce that is already hot. ASSISTED: the custard's batch in its pan, the temperatures of the food, the pan's base and its wall written once (food 80 to 95 C, plate and wall 100 C), the knob written into a detent, the flat stirrer carried along the custard's scraping path by a phantom carriage, or nothing stirring. The hob's own energy ledger is read every second.

Case	Into the pan	To the food	of it by boiling on the plate	Evaporated at the surface	Left as steam from boiling	The sauce heats	Burnt on
88 C, 1000 W, stirred, 20 s	840 W	747 W	381 W	147 W	0	0.30 C/s	nothing
88 C, 1800 W, stirred, 10 s	1512 W	1324 W	879 W	147 W	0	0.58 C/s	nothing
95 C, 1000 W, stirred, 45 s	840 W	765 W	521 W	224 W	378 W	0.08 C/s, to 98.6 C	nothing
80 C, 1000 W, left alone, 60 s	840 W	679 W	641 W	101 W	8 W	0.28 C/s, to 96.9 C	from 12 s; the whole bottom, 181 cm2, by 40 s

A higher setting heats a hot sauce faster (1.8 times the watts give 1.77 times the food's share), the vapour made at the pan's bottom condenses in the sauce while its mean is under the 96 C threshold (section 7), and a stirred pan does not catch. Left alone, the film on the bottom dries and burns on: the first patch at 12 s, past the task's 2 cm2 at once, and the whole bottom of the pan, 181 cm2, by 40 s. The burnt film is a layer the plate's heat has to pass through, and the plate under it runs hotter by that layer's own drop and then settles: its hottest patch 108 C at 10 s, 125 C at 20 s, 130 C from 40 s to the end (the plate's mean 122 C), while the sauce goes on heating, at 0.28 C/s on average, to 96.9 C at 60 s. The layer's conductivity is a dairy fouling deposit's, measured on a heated stainless surface (section 7); it does not thicken with time, and nothing chars further (section 12). From above, a pan whose bottom has burnt on looks like one that has not, as a real one under an opaque sauce does: what a policy has is the thermometer and the dark flecks that scraping a burnt patch brings up into the sauce. Stirred at 95 C the sauce goes to a simmer: its mean rises to 98.6 C, and on average 378 W of the vapour made at the bottom leaves it (section 7: above a mean of 96 C) rather than condensing in it. The surface's evaporation (147 W at a sauce near 88 C) is the measured pot's per area (section 7): Berick's 5 litres at 88 C lost about 600 g/h from a pot 25 cm across, which over this pan's 15 cm is about 220 g/h, 140 W.

Heating from cold is shown by the custard and porridge end states (section 11.6): 1000 W from 20 C, stirred, takes the custard to 93 C at 238 s with 62 % of the electricity in the food, and burns nothing on. A bench that heated the custard from cold, still and scraped, was run only on an earlier version of the heat model (section 12).

The probe thermometer (ASSISTED: a phantom carriage dips it into food written to 60 C). It reads 20 C in the crock, then 0.67, 0.88, 0.96 and 0.995 of the step 1, 2, 3 and 5 s after its tip enters the food, where a first-order sensor of 1 s gives 0.63, 0.86, 0.95 and 0.99; 59.8 C at the end of the dip, 46.9 C six seconds after it is lifted out. No rollout takes it from the crock.

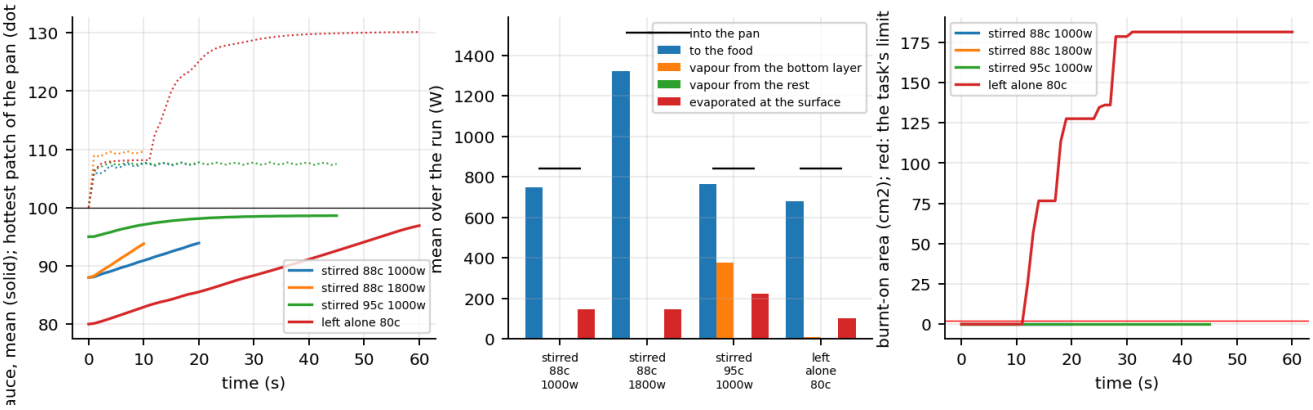


Figure 15. Hot-pan energy and temperature histories. A burnt patch passes heat; the plate settles near 130 C.

11.5 The robot does the tasks

ROBOT ACTIONS: a scripted operator, not a policy, drives the arms through the public action interface only (joint targets, the gripper, `set_load`); it reads the simulator's poses to plan its path, which a policy could not. Nothing is placed, held or moved for it. These runs show that the world lets a robot do the work and what the work costs it; they do not show that a policy can.

What a run records. A muffin rollout can pass with the bowl tipped 13.7 degrees on the way, so every rollout record carries `vessel_watch`, the vessel's tilt, lift off the counter, slide and turn read on every 1 ms tick, with the largest of each and when it happened. The operator's stroke stream waits for the arm (`keep_up` 0.02 rad, on for a Panda by default), and the Panda's fingers collide through the FR3's four finger boxes alone (section 9). Which runs had the stream paced is in the table.

Table 11.5a. The robot with the food. "Paced": the stroke stream waits for the arm. The vessel's peak tilt and the rise of its rim are the worst over every tick of the run; at the end the evaluator reads the vessel where it stands.

Rig	Task, strokes	Paced	The evaluator at the end	Vessel: peak tilt, rim lift	Stops, arm contact
dual-fr3	muffin-fold, 19 to the edge	yes	inside all five: 0.03 % dry, unevenness 0.035, 7.42 of 8.25 J/kg	2.4 degrees, 3.8 mm	0, 0
single-fr3	muffin-fold, 19 to the edge	no	inside all five: 0.0 % dry, 0.028, 7.79 J/kg; bowl 13.9 mm from its place	13.0 degrees, 20.4 mm	0, -
single-panda	muffin-fold, 19 to the edge	yes	inside all five: 0.16 % dry, 0.060, 7.43 J/kg; bowl 9.8 mm	10.7 degrees, 16.9 mm	0, -
dual-fr3	pancake-whisk, 44 to the edge	yes	NOT inside: unevenness 0.142 (0.10), no dry flour, 3.56 J/kg	0.02 degrees	0, 0
dual-fr3	pancake-whisk, 60 planned	no	NOT done: a protective stop on the first stroke; the whisk put back	0.01 degrees	1, 0
single-panda	sauce-blend, 70	yes	inside: the least even paste 0.089	0.01 degrees	0, -
single-fr3	sauce-blend, 70	no	NOT inside: the relish 0.107 (0.10), the ketchup 0.020	0.03 degrees	0, -
dual-panda	sauce-blend, 70, the left arm aside	yes	inside: the least even paste 0.089	0.01 degrees	0, 0
dual-panda	stir-sauce, 181, two-handed	yes	NOT inside: 110 cm2 burnt on (2); all the starch cooked, no lumps, knob at 0	0.0 degrees, pan held	0, 0

What the table shows. A robot folds a muffin batter to all five limits with either arm alone or with the pair, blends a sauce with a Panda alone and with the Panda pair, and cooks a custard with two hands on the hob, holding the pan, stirring and turning the knob, with no protective stop and no arm-to-arm contact. What it does not do, as scripted:

- *The pancake.* Paced, the strokes ran at half the plan's speed (the stream waited 47 s for the arm over 87 s of work) and 44 of them left the batter at 0.142, falling about 0.01 every four strokes, with 4.7 J/kg of the work allowance still unused; unpaced, at the plan's 0.40 m/s, the first stroke drove the whisk into the bowl and the arm's collision reflex stopped it (joint 2 at 22.4 N m against 20; the whisk pressed on the hand's flange at 123 N). The assisted carriage, at the plan's speed with no arm behind it, finishes in 38 s (section 11.6). No run of 70 paced strokes was made.
- *The sauce with the right FR3 alone.* Its 70 strokes, unpaced, end with the relish at 0.107 against 0.10; the ketchup alone reads 0.020 and would pass. The Panda rollouts, paced, lasted 170 and 172 s against the FR3's 153 s, and end at 0.089.

- *The custard.* The operator's strokes go through the middle of the pan in a new direction each time; the film under the sauce burns on where they do not reach, from 150 s (7 cm2) to 110 cm2 at the end. The assisted run's path scrapes the whole bottom and burns none. The world punishes a path that does not renew the film, as a pan does.
- *The bowl, with one arm.* Nobody holds it: under the right FR3 (unpaced) and the right Panda (paced) the bowl tipped 13.0 and 10.7 degrees in the strokes to the edge, its rim 20.4 and 16.9 mm off the counter, and settled; the FR3 pair, paced, 2.4 degrees. The evaluator judges where the bowl ends (it ended 13.9 and 9.8 mm from its place, inside the 50 mm); the record says what happened on the way.
- *The Panda's spatula.* With the stream paced and the fingers' meshes off, the right Panda's spatula read 2.8, 20.9 and 1.7 mm from its place in the hand at the ends of strokes 5, 10 and 15 (the blade at the bowl, its give included); the per-tick reading of the hand that separated the handle's turn from the blade's give was made for the dry strokes (section 9), not for this run.

11.6 Every task carried to its end state with assistance

ASSISTED. A phantom carriage carries the tool along a path fixed beforehand; the knob and the mixer's switch are written into their detents; no robot. It shows what the world allows a tool to do, not what a robot can do (section 11.5 has the robot). The evaluator is read every 2 s; a run ends when two marks in a row have every criterion of mixing met, or at its time limit; the tool is then lifted out (and the mixer switched off and set down, the hob turned off) and the evaluator is read once more. The batters start after a 13 s wait, the time the scripted robot needs to fetch its tool, during which the liquid soaks into the powder untouched.

Table 11.6a. Every task inside all its limits once, ASSISTED. "Inside from" is the first of the two marks in a row with every criterion of mixing met; the reading is the last one, the tool out and the appliance off.

Task	Tool and path	Inside from	The last reading (limit)
muffin-fold	spatula, strokes to the batter's edge, after the 13 s wait	22 s	flour unevenness 0.084 (0.10); dry flour 0.55 % (2 %); work 5.91 J/kg (8.25); nothing spilt; bowl moved 9.9 mm (50)
dry-whisk	whisk, brisk loops; three processes	98 s; 120 s; the third NOT inside at its end	leavening and salt 0.085, 0.081 and 0.104, sugar 0.034, 0.013 and 0.037 (0.10); spilt 0.59, 0.58 and 0.68 % (1 %)
pancake-whisk	whisk, strokes to the edge at 0.40 m/s, after the 13 s wait	38 s	flour 0.090; no dry flour; work 4.47 J/kg (8.25); nothing spilt; bowl unmoved
pudding-whisk	whisk, brisk loops, after the 13 s wait	44 s	0.042; no dry mix; lumps 1.79 % (2 %); whisk out
cake-mixer	hand mixer at setting 1, then 3, beaters 5 mm over the floor	30 s	0.041; no dry mix; lumps 1.46 % (2 %); mixer off, set down out of the bowl
sauce-blend	spoon, strokes	118 s	the least even paste, the relish, 0.078 (ketchup 0.064, mayonnaise 0.045)
stir-sauce	flat stirrer scraping the bottom; 1000 W, from 93 C 500 W for a minute	298 s of stirring	all the starch cooked (95 %), 96.8 C (88 C); no lumps; nothing burnt on (2 cm2); knob at 0
reheat-porridge	flat stirrer scraping the bottom; 1000 W, from 214 s 580 W	270 s	coldest portion 74.9 C (74 C), spread 6.2 K (15 K); nothing burnt on; knob at 0

`pancake-whisk`**, with the whisk.** Three paths, each strokes across the bowl in a new direction every time, 6 mm over the floor: to 80 mm from the axis at 0.25 m/s (`pancake_across`), the same to 100 mm (`pancake_to_the_edge`, the batter's edge), and to 100 mm at 0.40 m/s (`pancake_quick`: "gently and quickly", the recipe's words). `pancake_quick` ends inside all five limits after 40 s of strokes: the flour's unevenness 0.0897 against 0.10 (0.083 to 0.094 over the four cut frames), none of it dry, the batter's work 4.47 J/kg against the 8.25 allowed (13.6 of the 25 reference strokes), nothing spilt, the bowl where it stood. The two slower paths end at their 90 s limit with every limit met but the unevenness, 0.124 and 0.109, the flour still richer in the middle of the bowl than at its wall, with 4.7 and 4.3 J/kg of work taken. On `pancake_across` the whisk leaves 29.5, 15.5 and 10.7 % of the flour dry at 10, 16 and 20 s, and a spatula on the same strokes 22.6, 6.2 and 0.8 %. On the same strokes the whisk spreads the flour less than a blade. At 10 s of `pancake_across` the outermost of four rings of equal area holds 0.12 of the batch's flour share (the spatula 0.99); a faster stroke spreads it (`pancake_quick`: 0.20 at 10 s, 0.80 at the end).

A control: the whisk's cage and the spatula's blade carried on the same five strokes of `pancake_across` through the same start (2 s after the hand-over), on the tasks' 2.5 mm cells. The cage wets the flour at about three quarters of the blade's rate (dry share 0.80 to 0.50 in 6 s against 0.80 to 0.40), for 0.41 of the blade's work (0.57 against 1.38 J/kg), and leaves the flour in the middle of the bowl (outermost ring 0.04 of the mean against the blade's 0.82). The same pair on cells of 1.25 mm (3.37 million points; Table 11.6b): the heap starts drier there (0.875 of the solid dry at the hand-over against 0.800), and in the same 6 s the cage takes the dry share down by 0.246 and the blade by 0.370, two thirds of the blade's rate (three quarters on 2.5 mm cells). The liquid taken into the cage's cuts is 45.9 mL against 50.6 mL. What the cell still changes is

the work: the cage's falls from 0.57 to 0.27 J/kg on the finer cells while the blade's stays at 1.36 to 1.38, so the wires' drag on the batter depends on the cell (as it does in the markers bench, section 11.3). Without the carry and cut rules of section 4 the cage takes a third of the blade's rate on 2.5 mm cells.

Table 11.6b. The cage and the blade, the same five strokes through the pancake heap, 6 s.

Cells	Tool	Dry share, start to 6 s	Fallen by	Outermost ring at 6 s	Work (J/kg)	Liquid into cuts
2.5 mm	cage	0.800 to 0.504	0.296	0.04	0.57	50.6 mL
2.5 mm	blade	0.800 to 0.396	0.404	0.82	1.38	-
1.25 mm	cage	0.875 to 0.629	0.246	0.003	0.27	45.9 mL
1.25 mm	blade	0.875 to 0.504	0.370	0.62	1.36	-

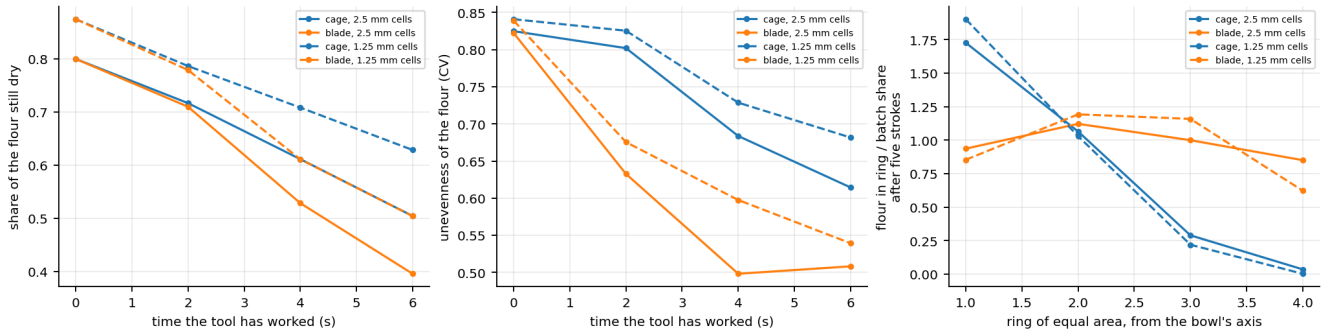


Figure 16. Cage and blade on the same five strokes through the pancake heap at two cell sizes. The cage's mixing work depends on the cell size.

`cake-mixer`**, the beaters at the floor.** The beaters' ends 5 mm over the floor in the flat middle and lifted by as much as the floor rises under the cages (the cages' round ends kept 5 mm off the bowl everywhere), the mixer taken round an ellipse 45 by 62 mm about the bowl's axis, drawn in to a quarter of it and out again every 7 s; the lowest setting until half the mix is moistened, then the third. Every reading says where the mix that is still dry lies, by height over the floor, radius and distance from the bowl's surface. At the start 76 % of the solid is dry; 19 % of that lies within 5 mm of the floor and 32 % between 20 and 40 mm. After 2 s on the lowest setting 26 % is dry, 40 % of it within 5 mm of the floor, and 213 mL of the liquid has gone into the cuts the beaters' strips leave. At 10 s (the third setting from 4 s) 1.8 % is dry, 91 % of it within 5 mm of the floor, under the beaters' ends; from 20 s none. Every criterion of mixing is met at 30 and 32 s, and read with the mixer switched off and set down out of the bowl: unevenness 0.041 against 0.10 (0.038 to 0.043 over the four cut frames), no dry mix, 1.46 % of the mix in lumps against 2 %, 0.005 % of the batch spilt, the bowl unmoved. The box's directions ask for about two and a half minutes (30 s on low, two minutes on medium); here the batter is combined in 32 s. What the two minutes on medium do to a real batter, beat air into it, is not simulated (section 12), and nothing here says how long a real batter takes to be combined.

`dry-whisk`**, with the whisk.** The brisk path (small fast loops, tilted, orbiting the bowl), judged on the leavening and salt, and the sugar, each over twelve portions of equal mass. Inside its limits after 100 s: the leavening 0.0854 against 0.10 (0.080 to 0.099 over the four cut frames), the sugar 0.034, 0.6 % of the batch on the counter against 1 %, the bowl 6.3 mm from where it stood. The reading moves between marks because the powder does: from 24 to 26 s it went from 1.08 to 1.23 as the whisk pushed a heap of leavening across a cut; the same state read again with its points shaken reads within 0.1 % of itself (section 11.8). Two further runs of the same path in fresh processes, with the same start to the last digit (2.0716): the solver gathers points onto the grid with atomic additions on the GPU, whose order is not fixed, so the runs part within seconds, and the leavening's reading differs between them by up to 0.35 at one mark (at 40 s 0.81, 0.58 and 0.64; at 60 s 0.60, 0.31 and 0.48) before they come together. They first meet the limit at 98, 108 and 108 s. The first is inside at 100 s (0.085) and the second at 122 s (0.081, having gone over again at 110 to 118 s); the third meets it at 108 and 110 s (0.097, 0.100) and reads 0.104 once the whisk is lifted out, and does not pass. So: the pass is shown twice in three runs, and where in the second minute the dry mix comes under the limit depends on the run: first crossings span ten seconds, while the two successful stops span 22 seconds and the third stop fails. A cook is said to blend these in about twenty strokes, ten seconds (section 12, limit 2).

`sauce-blend`**, with the spoon.** Strokes of the spoon across the bowl in a new direction each time, to 62 mm from the axis at 0.12 m/s, 4 mm over the floor, judged on all three pastes, the least even of them deciding. Inside the limit from 118 s of strokes (two marks in a row, 118 and 120 s); read with the spoon out at 120 s: the relish 0.078 against 0.10 (0.076 to 0.081 over the four cut frames), the ketchup 0.064, the mayonnaise 0.045, nothing spilt, the bowl where it stood. At 84 s the least even paste still read 0.186.

`pudding-whisk`**, with the whisk.** After the 13 s wait: the whisk in fast loops of 35 mm radius, 2.2 a second, tilted 25 degrees, carried round the bowl. Every criterion of mixing is met at 44 and 46 s of whisking; read with the whisk out: unevenness 0.042 against 0.10 (0.037 to 0.047 over the four cut frames), no dry mix, 1.79 % of the mix in lumps against 2 %, 0.001 % spilt, the bowl unmoved. The lumps were 2.28 % at 40 s and fell by 0.13 to 0.19 points a mark (2 s): the pass is narrow at the instant it is first met and widens with more whisking. The box says "whisk 2 min" and "soft-set in 5 min" (section 8): here the mix is combined in 46 s, and the setting goes on after the whisk is out.

`stir-sauce`**, with the flat stirrer.** The stirrer scraping the whole bottom, the hob at the task's 1000 W until the sauce is at 93 C (the maker's "bubbles are forming and bursting"), then the knob down and the stirring kept up "1 additional minute" (King Arthur Baking, pastry cream). 95 % of the starch is cooked at 234 s and the sauce's mean is 93 C at 238 s; the knob goes to 500 W at 240 s and to 0 after 298 s of stirring. Read with the stirrer out and the knob at 0 (303 s): all the starch cooked, no lumps, nothing burnt on, nothing spilt, the pan unmoved, the sauce at 96.8 C.

`reheat-porridge`**, with the flat stirrer.** The stirrer scraping the whole bottom; the knob to 1000 W at the start and back to the task's 580 W at 214 s. The coldest portion is first at 74 C at 270 s; the stirring stops there and the knob is turned off at 275 s. Read then: the coldest portion 74.9 C, the hottest 81.1 C (a spread of 6.2 K against the 15 K allowed), nothing burnt on, nothing spilt, the pan unmoved.

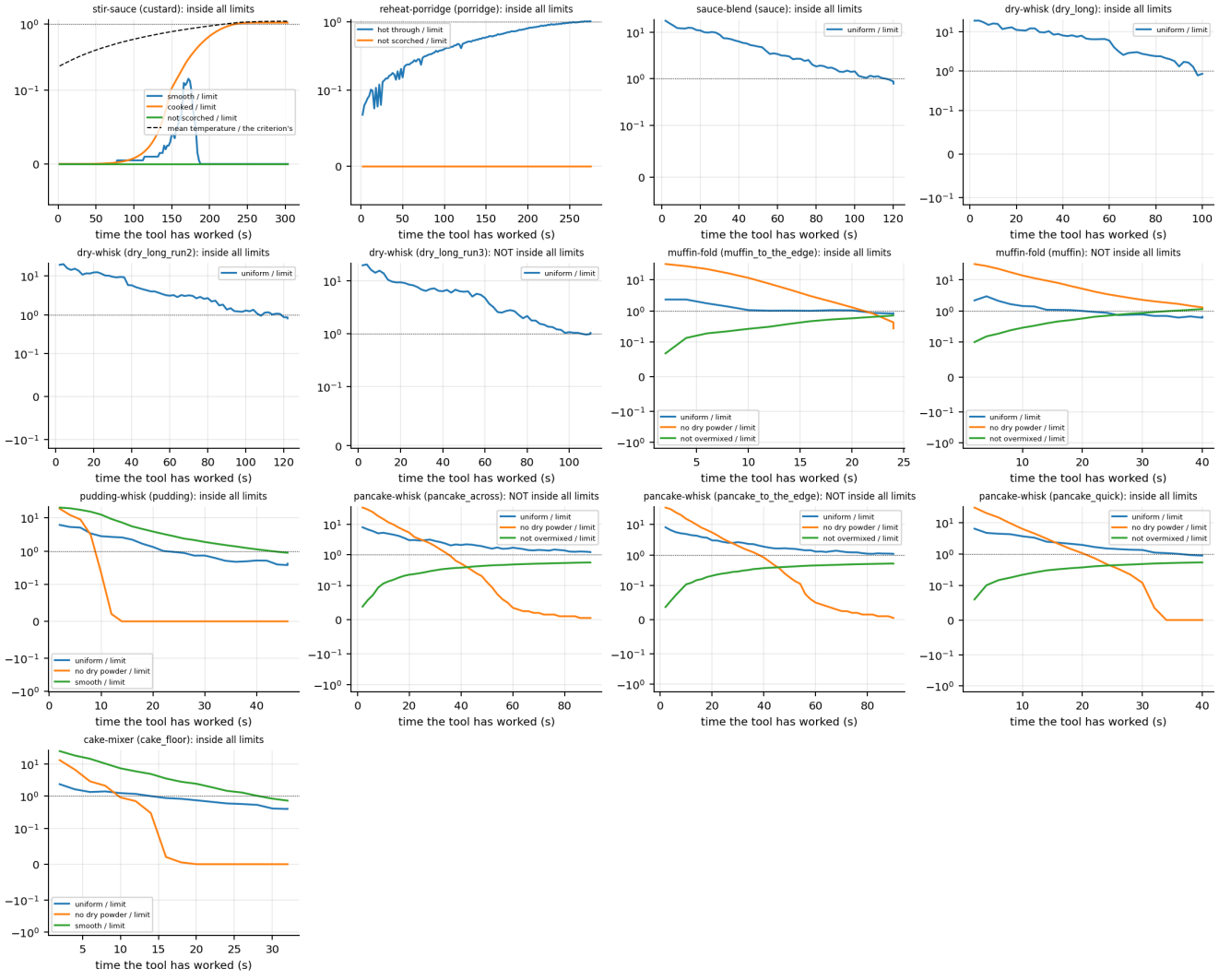


Figure 17. The assisted end-state runs. Every task has a passing record.

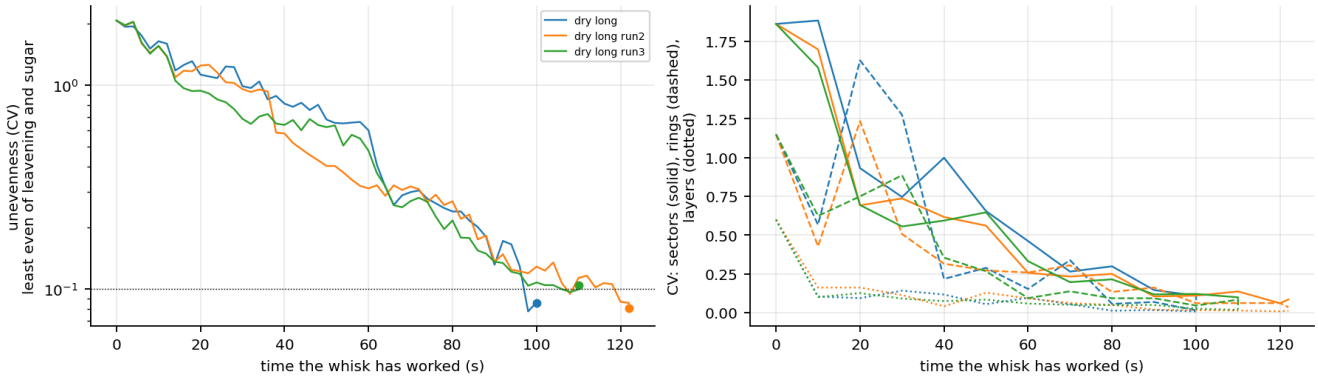


Figure 18. Dry-whisk histories from the three processes, including the third run that fails after the whisk is lifted.

11.7 The room, its furniture and what stands on the counter exist physically

ASSISTED: a spoon (in the hob scenes the stirrer) is placed in the air and dropped, thrown or tossed; there is no robot in this test. From either arm base in either preset a hand reaches 0.96 m and a tool held at full stretch 1.26 m; everything drawn inside that, and the furniture, the walls and the ceiling beyond it, has a collision proxy made from the same numbers as its drawing: static for the room, its fittings and its furniture, a free body for each thing on the counter. The probe aims the tool's end at a point on each one and records the first contact the tool makes. A thing that was only drawn would let the tool through to whatever is behind it, and the record would name that instead.

What	How	First contact	Where (m)	Aimed at (m)
sink top, right of the bowl and in front of it	dropped	its top slabs	z = 0.9135	0.914
sink bowl floor	dropped into the bowl	the unit's base	z = 0.7335	0.734
tap column, tap spout	dropped	each its own proxy	z = 1.2055	1.206
bottle on the sink	dropped	bottle	z = 1.1935	1.194
towel, rail rod, rail post	dropped	each its own proxy	z = 1.3095, 1.3055, 1.3045	1.310, 1.306, 1.305
canisters 0-2, jars 0-1, prep bowls	dropped	each its own proxy	z within 0.5 mm	
socket plates 0-1	thrown at the wall	the plate, not the wall	y = 0.6284	0.628 (wall at 0.635)
shelf bracket: wall plate, brace	pointed level, handle first	plate (y = 0.6301), brace		0.631
shelf bracket: arm	poked up beside the brace	the arm	z = 1.7355	1.736
light strip under the shelf	tossed up	the strip	z = 1.7321	1.732
station camera on its bracket	poked up from below	its housing	z = 1.5758	1.576
prep table, two stools, storage rack	dropped on their tops	each its own proxy	z = 0.8995, 0.7495, 0.7495, 2.1395	0.90, 0.75, 0.75, 2.14
hand-wash basin, its tap, towel dispenser, pedal bin	dropped	each its own proxy	z = 0.8595, 0.9895, 1.6195, 0.6195	0.86, 0.99, 1.62, 0.62
fridge: top, upper door, side	dropped on the top; thrown at the door and at the side	the cabinet, the door, the cabinet	z = 1.7995; y = -0.0317; x = -1.6708	1.80; -0.032; -1.67
plant's pot: rim	dropped	the pot's topmost cylinder	z = 0.3419	0.341
front wall, left wall, right wall (the window's), ceiling	thrown	the wall	y = -3.5653, x = -2.4008, x = 3.0007, z = 2.6998	-3.565, -2.4, 3.0, 2.7
what stands on the counter in the eight tasks' scenes: 39 things, each a free body	the low ones (egg carton, scale, tub, towel, pudding box): dropped on their top; the tall ones: the tool, lying level, pointed handle first at their front	each its own body	on its top, or on the front face of its outline as it is turned	

77 of 77 (`passed`, `of`): The 39 things on the counter were all struck (`free_bodies_struck`); 13 of them were pushed by more than half a millimetre, none was tipped, and all 39 were at rest 1.5 s afterwards (`free_bodies_moved`, `free_bodies_at_rest_after`, and `free_body_then` on each row). A thrown 32 g spoon pushes a 0.28 kg baking tin 4.5 mm and a 0.43 kg starch box 4.6 mm, a 0.30 kg salt 3.4 mm and a 0.45 kg jug 0.5 to 3.3 mm; a 1 kg bag of flour or a carton of milk moves 0.0 to 0.5 mm. This probe throws a light tool; it does not show an arm sweeping a pack over, and no run of the code as it is does (the careless commands of section 11.8 were run on an earlier version).

What is still fixed: the walls, the furniture and the fittings. A canister cannot be knocked off the shelf, a stool cannot be pushed, a round fitting is the square around it, the towel on its rail is a rigid slab, and the window is a wall. The fridge is its boxes and the plant's pot its stacked cylinders; the plant's leaves have no collider.

Props where they were placed: all 39 packs of the eight tasks' scenes are where they were placed one second after the build, each within 0.5 mm and 0.5 degrees and slower than 0.05 mm/s (`props` 39, `not_where_placed` 0).

11.8 The whole environment

The evaluator against states with known answers.

ASSISTED: each state is written into the simulator (compositions, temperatures, flags, poses, joints), so that the right verdict is known before the evaluator reads it. 42 pairs, all 42 agree (`all_agree`).

Among them: in the cold sauce, the ketchup even and one quarter of the bowl holding half of the other quarters' relish (0.78, not passed; judged on the ketchup alone this state would pass); in the dry mix, the rest even and one quarter holding half of the other quarters' sugar, and the same for the baking powder and salt (0.70 each, not passed); and the dry mix's start state read five times with every point shaken by a fifth of the point spacing, which reads 2.0439 to 2.0453 against 2.0458 unshaken, within 0.1 % of itself. The other pairs: the sauce and the dry mix at their starts and at the batch composition everywhere, spilt shares on either side of the 1 % limit, a bowl moved 30 and 70 mm, dry flour at 1 and 4 %, work of 20 and 30 reference strokes, lumps at 1 and 4 %, the whisk in the crock and left in the bowl, the mixer switched off and left running, starch cooked to 90 and 97 % at 80 and 88 C, cooked starch even and in a 12 mm clump, burnt patches of 1.65, 3.31 and 3.54 cm2, the knob at 0 and left at 4, porridge at 72 and 76 C everywhere, with a cold core of 60 mm and with a hot core smaller and larger than a portion.

A start state built as a bench builds it and as a session does. Built both ways and compared at the hand-over, the two states are the same: no point is more than 0.07 micrometres (dry mix) or 0.24 micrometres (sauce) from its place in the other, and both read 2.0458 (dry mix) and 1.7292 (sauce). A bench reads its row `start` later, after the tool has been brought into the vessel and, for the batters, after the 13 s wait, so the end-state runs also record `as_built`, the reading at the hand-over.

Every start state left alone (NOTHING ACTS). Each of the eight start states is built as a session builds it and left 21 s with the evaluator read every 3 s. None passes. The four that put a liquid against dry powder go on wetting with nobody touching them: between 3 and 21 s the dry share of the solid falls from 0.74 to 0.66 (`muffin-fold`), 0.81 to 0.74 (`pancake-whisk`), 0.67 to 0.57 (`pudding-whisk`) and 0.80 to 0.75 (`cake-mixer`), and their points move 0.17 to 0.75 mm at the median; their 99th-percentile speed is 0.07 to 0.13 mm/s at 21 s. The dry mix lies still (3 micrometres at the 99th percentile in 21 s); the cold sauce's spoonfuls spread (0.04 mm at the median, 3.2 mm at the 99th percentile); the pan's cold slurry drifts (0.18 mm at the median, 0.84 at the 99th percentile, 2.4 mm at most, its 99th-percentile speed holding at 0.05 mm/s); the cold porridge moves 0.03 mm at the median and 0.48 mm at the 99th percentile. No pack, jar or tool on the counter moved.

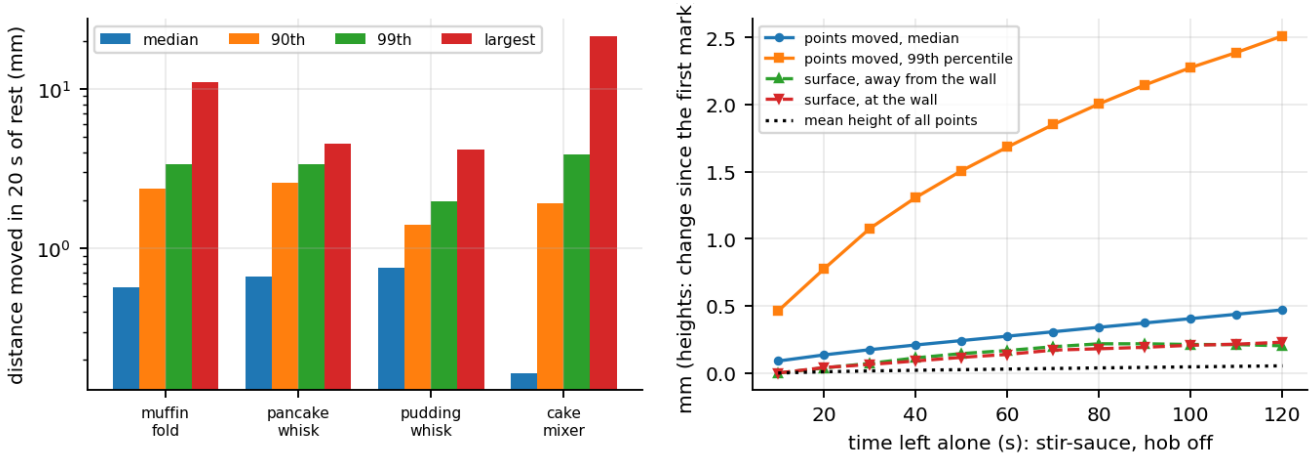


Figure 19. Start states left alone, showing motion and continued wetting before any tool acts.

Careless commands (LEGAL ACTIONS, `pancake-whisk`; run on an earlier version of the code): 35.6 s of rough commands a policy may send, nothing assisted: full-speed joint moves, the closed hand plunged into the batter and swept through it at the keyboard's fastest speed, the bowl pushed by the hand until it slides, the crock knocked, the hand opened and closed at full force on nothing, a tool taken and dropped into the bowl from 0.2 m. The solver stays finite with no fault and no hit on its speed guard; the arms' reflexes stop them twice and two commands are refused; the crock is knocked over (tipped 86 degrees, 305 mm from its place), the three tools are thrown 0.52 to 0.64 m, the bowl ends 44 mm from its place, tilted; the packs on the counter do not move. Batch mass is kept to 6e-6 of itself.

Run time through the gateway. Sessions started as a policy gets one (`gar run ... --policy-key`), driven over HTTP with the restricted key, every image downloaded inside the timed interval; each command's record carries the machine's load as the command was sent (the card's use and memory, its processes and whose they are, the CPU's pressure). Ratio: wall seconds from the request to the last image byte over simulated seconds; the limit is 200.

Case	Count	Median	Worst	Others' processes on the card
`sauce-blend`, no camera	21	16.8	59.6	2
`cake-mixer`, no camera	21	39.4	99.0	3
`sauce-blend`, one camera	16	27.3	38.3	3
`sauce-blend`, four cameras	12	33.5	35.3	3
`cake-mixer`, one camera	29	49.6	68.7	3 to 6
`muffin-fold`, four cameras	16	86.5	99.3	4 to 6
`stir-sauce`, one camera, the hob on, after a short release	18	94.6	130.6	5 to 7
`sauce-blend`, one camera, after the 15-minute release	8	62.0	122.5	3 to 6
`sauce-blend`, one camera, the render worker killed	11	40.1	414.6 (the command under which it died; told so)	1 to 5
three sessions at once (`cake-mixer`, `muffin-fold`, `stir-sauce`)	42	113 to 174	1347, 10086, 8962	1 to 5, with four render workers of another session

Every command of a single session is inside 200x but the one under which its render worker was killed, which says so (`over_budget` 1). Three sessions at once are not: with another session's four render workers on the card as well (18 processes, 31.3 of 32.6 GB in use), four of their commands took 480 to 10,086 times real time (a 0.064 s command 645 s, nearly all of it the physics waiting for the card), and every one of the four told the policy so in its reply; their other 38 commands are inside. A session cannot hold its budget on a card other sessions fill; it can and does say when it has not. On a quiet card an earlier version of the code had three sessions at once inside 200x.

The render worker's life (the release shortened to 6 s and 600 s for the test). A command whose physics outlasts the release keeps its worker; the worker is released after a pause and the floor rises from 0.064 to 0.123 s to cover a new worker's start, and the first command after it runs at 100.9x; two observations slowed by 15 s (ASSISTED: a sleep) raise the floor to 0.095 s, a command of the published 0.064 s is then refused, the shortest the raised floor allows runs at 180.4x where one of 0.064 s would have run at 261.6x, and three normal observations bring the floor back to 0.064 s. With the worker's process killed under a session, a command of the old floor's length is refused (the floor in force is 0.127 s), the command at that floor gets its image from a worker started beside its physics (93.1x), the next runs at 16.5x; an observation

with no command before it starts its own worker (12.3 s); a worker whose socket is shut from the session's end is replaced inside the command (72.2x) and the old process is reaped. No worker process is left after the session closes.

A live session as a user and as a policy. `gar inspect` lists the four robot choices; a launch with the left FR3 alone, a misspelt or unknown renderer option, 100,000 samples a pixel or a negative idle release is refused before anything is built, each with its reason; `gar run` starts a `sauce-blend` session in 52 s; `status`, `observe`, `verify`, `action --help-actions`, `action`, `dashboard` and `stop` work. With the restricted key the policy reads its capabilities and the task, gets images, is refused the owner's routes (403) and is refused, before anything runs, a command shorter than the floor, a grip of 200 N and a pose two metres up, and has a Cartesian move, a grip and a keyboard chord committed. The owner resets to the same task (1.6 s), is refused a reset to an unknown task, recovers, and resets to another seed (the world rebuilt, 94 s). The dashboard, opened with each key, shows the task, the policy's camera and the action form, the key field cleared (Figure 7). On a card full of other sessions' render workers the same session took 39 minutes to start and its reset to a new seed timed out at 30 minutes.

Views. 56 stills were rendered, each recorded with its task, camera, resolution and render cost. Figures 1 to 8 use them; a start-state still is not evidence of a task's outcome.

12 / Limits

Stated plainly. Each is something the real activity has and this environment does not, something it does that the real one would not, or something that was not measured. Numbers are from the records of section 11.

The tasks and the robot

1. *Every task is shown inside all its limits with assistance; the robot has not done every task.* ASSISTED (a phantom carriage carries the tool): eight of eight, each once and `dry-whisk` in two of three processes (Table 11.6a). By the robot with the food (a scripted operator, not a policy; section 11.5): `muffin-fold` inside all limits by the FR3 pair, the right FR3 alone and the right Panda alone; `sauce-blend` by the right Panda alone and the Panda pair, and by the right FR3 alone NOT inside (the relish 0.107); `pancake-whisk` by the FR3 pair NOT inside (unevenness 0.142 after 44 strokes, the stream paced) and stopped by a protective stop on the first stroke unpaced; `stir-sauce` by the Panda pair carried out and NOT inside (110 cm2 burnt on under strokes through the middle of the pan). `dry-whisk`, `pudding-whisk`, `cake-mixer` and `reheat-porridge` were not run by the robot with the food on the code as it is (a mixer rollout exists only from an earlier version).

2. *The whisk on dry powder is slow, and where it meets the limit depends on the run.* Three processes of the same brisk path read the leavening and salt at 1.64, 1.55 and 1.56 at 10 s, 0.99, 0.96 and 0.68 at 30 s, 0.60, 0.31 and 0.48 at 60 s, and first meet the 0.10 limit at 98, 108 and 108 s; two are inside at their end, the third reads 0.104 with the whisk lifted out. A baker's "approximately 20 strokes" (The Cake Blog) is about ten seconds. In the bench it was fitted on the solver's wire carries four fifths of what a resolved wire does; a gap of ten times to a cook is not that. Not measured: whether whisked flour aerates and flows (it does not here: a powder is a Drucker-Prager continuum with cohesion, and nothing fluffs it), and how much of a cook's blending happens below the points' 1.25 mm spacing (here only the grains' shear diffusion, section 5).

3. *The rules for a wire stand on one resolved wire, and the cell still shows where they were not fitted.* The carry rule was read at 0.25 m/s from an upright wire, and there the solver's wire carries four fifths of the resolved wire's powder on both cell sizes (8.4 and 8.1 against 10.4 to 10.5 mm2 per length). Away from that case it falls short and the grid's own share comes back: at 0.60 m/s 8.4 and 12.8 against 14.1; lying 10 mm under the powder 8.4 and 10.3 against 22.6 to 23.2; two wires three diameters apart 19.2 and 10.1 a wire against 21.1, a factor of 1.9 between the two cell sizes. The liquid's dose into a cut is the bed's (3.05 wire diameters squared, the rule 3.0); a small or loose heap takes more, a 1 Pa s liquid a quarter more (3.93), and a beater's 4 mm strip is held to a round wire of its width (2.63 at 0.6 m/s); no flat strip was resolved (section 11.3). The wire's drag in dry flour is below the resolved wire's (1.8 and 2.9 N/m on the two cells against 3.7 to 3.8): the load a held whisk feels in powder is low by that much, and depends on the cell.

4. *Robots.* Two templates only: two fixed-base arms, one fixed-base arm, each Franka FR3 or Panda. The humanoid is not supported. One arm alone cannot steady the bowl: in the muffin strokes to the edge the bowl, held by nobody, tipped 13.0 degrees with its rim 20.4 mm off the counter under the right FR3 and 10.7 degrees and 16.9 mm under the right Panda (past 2 degrees for 564 and 694 ms in all), and settled again (the FR3 pair, paced, 2.4 degrees). The Panda's spatula read 20.9 mm off its place in the hand at one of three marks of that run (2.8 and 1.7 at the others, read with the blade at the bowl). The Panda pair's custard rehearsal without the food ends with one protective stop. The FR3 pair lays the mixer down after the cake rather than standing it on its heel (rehearsal). The operator streams the FR3's strokes unpaced unless told (`--keep-up`); paced, the pancake strokes took twice as long as planned.

The mixture

1. *Thin liquids are too thick, and creep at rest.* In the stirred cup the torque is 1.35 times the laminar result at 0.2 Pa s and 4.2 times at 0.01 Pa s (milk is 0.002); a thin liquid left alone in the cup moves 0.59 mm at the median in 12 s and 5.7 mm in the corner of floor and wall (section 11.1). The pan's own start, the cold milk and starch of `stir-sauce` left alone, has moved 0.18 mm at the median, 0.84 mm at the 99th percentile and 2.4 mm at most by 21 s, its 99th-percentile speed holding at 0.05 mm/s: it does not stop. A jerked bowl of milk loses its swing at 5.4 to 13.4 % of critical damping where water has about half a per cent (run on an earlier version of the solver). The force behind the creep was found (section 4: the pressure's thrust on a wall node gathered from one side has a part along the wall) and taking it out stopped the creep at the walls and broke the free surface and the stirred cup's torque balance: not cured. A rest stress of 1 Pa on every wet state is what lets a thin liquid come to rest at all.

2. *A solid is one cell fat.* Material approaching a tool or a wall stops one cell (2.5 mm) short of it; a spatula's 5 mm blade pushes like a 10 mm one; the spoon's lip is 5 mm where a real one is 3 to 4 mm.

3. *Mixing below 2.5 mm is a closure* (section 5), with borrowed constants. The unevenness is read on portions of 5 mL and more, where the resolved flow decides.

4. *Wetting, lumps, hydration, setting and over-mixing follow first-order laws whose constants are not measured.* The

wetting constant is calibrated on the muffin method's stroke count; no strength or erosion rate of a wetted flour lump was found anywhere (the one paper on the topic could not be opened). The pudding's and the cake mix's rates are not set on their boxes' directions (two minutes of whisking; low speed 30 s, then medium two minutes): here the pudding is combined in 46 s and the cake batter in 32 s.

5. *The film on wires and beaters is a model with no measurement behind it.* Switched on at its top speed in the unmixed cake batter, the mixer throws batter as high as 128 mm over the bowl's floor, 14 mm over the rim, and none over it; how much a real hand mixer throws at that start was not found anywhere.

6. *No air.* Nothing foams, creams or rises; powder does not puff when liquid hits it. Steam and bubbles are drawn, not simulated. A powder that has been thrown stays fluffed.

7. *Spilt material is dead.* Where it lands outside the grid's region it sticks and stays; it is drawn as a puddle no deeper than 6 mm.

8. *Start states are put in, not poured* (section 10).

Heat

1. The food is heated through a film model on 96 patches of the pan's base, with a natural-convection closure for liquids below the solver's floor. The film's thickness (0.15 mm) that sets how soon an unstirred sauce catches is an estimate; no scorch time was found to set it on. A burnt-on film is a layer of a fouling deposit's conductivity grown in a heated tube, transferred to a scorched starch film; it does not thicken with time and nothing chars further. The evaporation and the vapour threshold are a measured pot of water, used for a sauce whose surface skins over. No skin forms and nothing boils over. A pan whose bottom has burnt on looks, from above, like one that has not (as a real one under an opaque sauce does; the smell and the sound are not simulated): a policy has the thermometer and the flecks that scraping brings up.

2. The hob is a knob and a zone; the real unit has keys, a display and a timer. The knob's detent is a maker's figure for a detented control, not this hob's.

3. The thermometer is shown in a carriage's hold, not in a robot's hand.

The room and the evidence

1. The room's furniture and fittings are fixed; what stands on the counter is not.

2. Nothing here was compared with a physical trial. The references are closed-form flows, published measurements of materials and appliances, and makers' directions.

3. *Run time depends on who else is on the card.* Through the gateway a single session keeps every ordinary command inside 200 times real time in the measured single-session cases; the injected worker-death command cost 414.6x. Three sessions at once, on a card another session's four render workers also held, had four commands at 480 to 10,086 times, each told so in its reply (section 11.8). The recorded simulations ran at 100 to 540 wall seconds per simulated second each beside up to six other sessions' render workers, and ten times slower again beside a render from the same job.

4. *Some records are of an earlier version of the code.* The custard heated from cold still and scraped, the muffin method's twenty strokes and a minute's standing, the rest variants, the thin liquid in a box, the tipped vessels, the jerked bowl of milk, the wall-seep benches, the three calibrations (the gate of section 11.1 reads the same at the fitted rows), the first wire benches, two beater variants and the robot's mixer rollout with the older robot runs were made before the wire's carry and cut, the liquid into cuts, the evaluator's portions, the hob's burnt layer, the evaporation and vapour threshold, the mixer's speeds, the knob's detent and the Panda's fingers took their present form. That the benches without any of these read as before is an inference, not a measurement. The mixer rollout and the custard heated from cold touch them.

13 / Reproduction

The environment is the package `ingredient_mixing`, started and driven with the `gar` command. A seed and a variant are selectors (`--select`); there is no `--seed` option:

```
gar inspect ingredient-mixing ; gar tasks inspect ingredient-mixing ; gar doctor ingredient-mixing
gar tasks resolve ingredient-mixing --select variant=honey          # which task, seed and text a selection gives
gar run ingredient-mixing --task muffin-fold --select seed=0        # prints the session, its URL and its keys
gar run ingredient-mixing --task sauce-blend --select seed=3 --select variant=honey --robot single-fr3 --port 0 \
    --policy-key
gar run ingredient-mixing --task dry-whisk --backend-options-json '{"cameras": []}'     # a session without cameras
gar observe SESSION ; gar verify SESSION ; gar dashboard SESSION ; gar stop SESSION
gar action SESSION --timeout 600 --idempotency-key hold-1 --data \
    '{"type": "named_joint_target", "targets_si": {}, "ticks": 10000}'
python -m ingredient_mixing.params              # every constant with its label and basis
```

On a card shared with other jobs a mixture run costs one to three minutes of wall time per simulated second, and a run of a few minutes of simulated time takes hours (section 12, limit 18); runs side by side slow each other, and a render worker beside them (a film, stills, a session's cameras) slows them about tenfold.

14 / References

The sources the numbers of this report rest on, grouped by how each was accessed.

Read. Lorente-Bailo S., Etayo I., Salvador M. L., Ferrer-Mairal A., Martínez M. A., Calvo B., Grasa J., "Modeling domestic pancake cooking incorporating the rheological properties of the batter. Application to seven batter recipes", preprint, 2020 (Zaguan record 106619): the wheat and spelt pancake batters' K and n, density, heat capacity, the rise of viscosity on heating, the hob's power. Nagasawa K., Suzuki T., Seto R., Okada M., Yue Y., "Mixing sauces: a viscosity blending model for shear thinning fluids", ACM Transactions on Graphics 38(4), 2019 (read in the authors' IPSJ summary): the blending rule. KitchenAid, Hand Mixer use and care guide W10586270C: the speed guide and the order of operations. Duxtop, portable induction cooktop BT-200DZ / 9600LS / 9610LS user manual: the overheat limit ("E2: Cooking surface temperature exceeds 460 degrees F (238 C) limit and the unit shuts down automatically"), pan detection, the sensor's place under the glass, "Output: 100-1800 watts" and the twenty power levels 0.5 to 10 with their wattages. Basic Scientific Food Preparation Lab Manual, 12.01 Muffin Method (LibreTexts); King Arthur Baking, Simply Perfect Pancakes; America's Test Kitchen, Stabilizing the Whisking Bowl: the recipes' proportions and the methods' words. Quaker Oats, Baking 101, Mixing muffin batter: "mix just until the dry ingredients are moistened, 15 to 20 light strokes" (the wetting constant's anchor and the over-mixing limit). King Arthur Baking, Simply Perfect Pancakes: "Stir into the milk/egg. Don't beat; some lumps are OK" (`pancake-whisk`'s criteria). America's Test Kitchen, on sifting: "Whisking is simply a way to fully combine your dry ingredients." Kansas State University Extension bulletin MF3393 (Stark and Saensukjaroenphon 2017), quoting Herrman and Behnke 1994: a mixer test's coefficient of variation under 10 % is excellent (the unevenness limit); MicroTracers, "Mix With Confidence" (1994) on what a 10 % limit means against the analysis's own variability; Pharmaceutical Technology on the compendial 6 % for content uniformity. gov.uk, food hygiene for businesses, cooking: 70 C for two minutes and its equivalents (the porridge's limit is the United States' 165 F, which was not opened). KitchenAid's specification sheet for the KHM5AP hand mixer: net weight 2.3 lb, five speeds, a heel rest, no wattage and no beater speeds. Liu, Christian, Zhang and Fryer, International Dairy Journal 16(2):164-172, 2006 (abstract): apparent adhesive strength of whey-protein fouling deposits on stainless steel, 0 to 10 J/m2 (the work of scraping a burnt-on film). Alharthi, PhD thesis, University of Birmingham, 2014: milk fouling conditions. The Engineering ToolBox's table of food bulk densities: flour 600 and granulated sugar 800 kg/m3. franka.de, Franka Hand: 730 g, a two-finger parallel gripper (nothing on its pads or on how its fingers are coupled). unitree.com, G1: 1.32 m standing, about 35 kg.

- A. Berick, "Heat Losses In A Cook Pot at Constant Temperature", Aprovecho Research Center (undated technical note): evaporation from 5 litres of water in a 25 cm stainless pot held at 50 to 98 C (figure 5) and the heating curve of the same pot under a film of oil (figure 7): `HOB_EVAPORATION`, `HOB_SIMMER_C`.
- F. J. Cantarero Rivera, D. K. Mishra, F. Ozadali, P. Benyathiar and others, "Real-Time Detection of Fouling-Layer with a Non-Intrusive Continuous Sensor (NICS) during Thermal Processing in Food Manufacturing", Sensors 21(4):1271, 2021, doi:10.3390/s21041271, full text through Europe PMC (PMC7916806): a dairy fouling layer's conductivity, `HOB_BURNT_K`.
- America's Test Kitchen, "Hand Mixers" equipment review: the KitchenAid 5-Speed Ultra Power hand mixer (KHM512) at 288 rpm lowest and 1,107 rpm highest, beater head 1.75 in: `MIXER_SPEEDS`, `MIXER_BEATER_R`.
- KitchenAid CLASSIC 5-speed Hand Mixer 5KHM5110 specification sheet, 2020: 85 W, DC motor: `MIXER_STALL` (an estimate from it).
- Multicomp slide switch, top actuating, data sheet V1.0 of 17/09/12: operating force 250 +/- 100 g: `MIXER_DETENT_FORCE`.
- Piher PTC-15 rotary position sensor and control data sheet: detent torque 0.5 to 3.5 N cm for more than 13 detents: `HOB_DETENT_NM`.
- A. A. Al-Tabbakh, "Worked Examples in Boiling and Condensation" and "Boiling Heat Transfer Coefficients Calculations" (lecture 2, 2021), Al-Mustansiriyah University; Majid, "Boiling-2" (course worked examples): water's properties and Rohsenow's surface constant on polished stainless steel, `HOB_BOIL_B`.
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Read second-hand, from research notes with quotes. King Arthur Baking, Pastry Cream: "The pastry cream should reach 190 F to 200 F on a digital thermometer", and, once it bubbles, "continue whisking for 1 additional minute" (`stir-sauce`'s 88 C). Sweeney, Dols, Fortenbery and Sharp, "Induction Cooking Technology Design and Assessment", ACEEE Summer Study 2014, table 2: two induction cookers put 74.9 to 77.6 % of the electricity into water heated from 70 to 200 F (the outcome the hob's efficiency is held against). ENERGY STAR, 2021 Residential Induction Cooking Tops: "approximately 85% efficiency" (the efficiency itself, 0.84). nuclear-power.com, Subcooled Boiling: "bubbles formed at the surface may condense in the

liquid". A worked example of Rohsenow's correlation, for water on brass. Tasting Table, quoting Pastry Chef Online: milk has "a grace period of 15 seconds at 180 degrees Fahrenheit" (the one figure found for how long a hot milk sauce may be left alone; a kitchen claim). A Massey University thesis on milk fouling of a tubular heat exchanger, table 4.4: dry foulant from 0.008 kg/m2 at a surface of 68 C to 0.296 kg/m2 at 93 C (fouling rises steeply with wall temperature; not used for a constant). Institute of Child Nutrition, Reheating Foods, and Government of Nunavut, Leftover Food Factsheet: 165 F, 74 C (the porridge's limit). Transcriptions of two cake-mix boxes: "Beat 2 minutes at medium speed (or 450 hand strokes)" (Duncan Hines), "on low speed for 30 seconds, then on medium speed for 2 minutes" (Betty Crocker). King Arthur Baking, Don't Make This Pancake Mistake: "a mere count of 10 strokes until the batter was just as it should be". The Cake Blog, Sifting for a Better Cake: "The cocoa is evenly and uniformly spread through the flour after approximately 20 strokes of the whisk". KitchenAid hand mixer owner's manual (KHM512 family): "should always be set on the lowest speed to start mixing", speed 1 "Helps avoid ingredient splash out". Engineering ToolBox, Heat Loss from Open Water Tanks: evaporation loss in still air, 1380 Btu/ft2h at 170 F to 2680 at 200 F (not used for the evaporation law). Ozuna et al., Foods 9(7):895, 2020 (muffin batter, 3.35 Pa s at 60 1/s); Flamminii et al., 2020 (mayonnaise: 78.8 Pa, 71.3 Pa s^0.3); Shelke et al., 1990 (layer cake batter, 1.95 Pa s); Huppertz and Nieuwenhuijse, International Dairy Journal 126:105236, 2022 (milk); product pages for the Vollrath 69050 bowl, the OXO 11.5 in whisk, the Choice 12 in spoon; the Duxtop 9100MC review (centurylife.org); King Arthur ingredient weight chart.

Looked for and not found. A time to scorch or a deposit thickness for an unstirred starch-and-milk sauce; a time constant or a strength for a wet flour lump; any study of powder blending by a wire whisk, or a number of seconds for whisking dry ingredients; a tolerance for lumps in a batter; friction coefficients for the pairs used here; the Franka Hand's pad material, friction or stiffness; the G1's wrist torque.

Seen in search summaries only. All-Clad D3 1.5 qt saucepan dimensions; NKBA counter figures; Jell-O and Betty Crocker package directions (secondary reproductions of the box copy only); the United States' 165 F for reheated food (the official page refused the fetch); the instant-read thermometers' response times; wheat flour's particle density; Rwei, Manas-Zloczower and Feke, Polymer Engineering and Science 31:558-562, 1991 (agglomerate erosion); Christianson and Bagley, Cereal Chemistry 60:116-121, 1983.

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This repository. The Franka FR3 model, limits and controller profile of `robots/` (qualified in `barcode_scanning`); `drilling` (the arm controller, the film layout), `wine_pouring` (the render worker, the lumped-contents coupling, the surface reconstruction).